

Integrating Human Factors into Insider Threat Detection – A Systematic Review

NADEEKA PATHIRANA, School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, United Kingdom of Great Britain and Northern Ireland

RUBY ROBERTS, Aberdeen Business School, Robert Gordon University, Aberdeen, United Kingdom of Great Britain and Northern Ireland

HARSHA KALUTARAGE, School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, United Kingdom of Great Britain and Northern Ireland

CHRISTOPHER D. MCDERMOTT, School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, United Kingdom of Great Britain and Northern Ireland

Insider threats, originating from current or former employees, contractors and trusted business partners who are with authorised access to organisational resources continue to pose a critical security challenge across multiple sectors. These threats may be intentional (malicious) or unintentional (accidental) and are shaped by the interaction between technical and human factors. This systematic literature review analyses 121 studies published between 2014 and October 2025 to address two key research questions: (1) which human factors contribute to insider threats, and (2) whether existing technical methods, particularly machine learning approaches, have been applied to predict or detect insider threats on the basis of human factors. The review introduces a taxonomy of human factors influencing insider behaviour and provides an examination of technical approaches, including detection methods. The findings highlight a substantial gap in machine learning research directly linking human factors to insider threat detection, largely due to the absence of datasets that integrate behavioural and contextual human attributes. By synthesising current knowledge and identifying these limitations, this study outlines key challenges and research opportunities, underscoring the need for predictive models that capture the complex role of human factors in insider threats especially in unintentional insider threats.

CCS Concepts: • **Human-centered computing** → Accessibility technologies; • **Security and privacy** → **Human and societal aspects of security and privacy**; *Social aspects of security and privacy*; Phishing; • **Computing methodologies** → **Artificial intelligence**; **Machine learning**;

Additional Key Words and Phrases: Insider threats, unintentional insider threats, human aspects of insider threats, technical aspects of insider threats

Authors' Contact Information: Nadeeka Pathirana (corresponding author), School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, Scotland, United Kingdom of Great Britain and Northern Ireland; e-mail: n.withana-pathirannahalage1@rgu.ac.uk; Ruby Roberts, Aberdeen Business School, Robert Gordon University, Aberdeen, Scotland, United Kingdom of Great Britain and Northern Ireland; e-mail: r.roberts2@rgu.ac.uk; Harsha Kalutarage, School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, United Kingdom of Great Britain and Northern Ireland; e-mail: h.kalutarage@rgu.ac.uk; Christopher D. McDermott, School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, Scotland, United Kingdom of Great Britain and Northern Ireland; e-mail: c.d.mcdermott@rgu.ac.uk.



This work is licensed under a Creative Commons Attribution 4.0 International License.

© 2026 Copyright held by the owner/author(s).

ACM 0360-0300/2026/04-ART260

<https://doi.org/10.1145/3798089>

ACM Reference Format:

Nadeeka Pathirana, Ruby Roberts, Harsha Kalutarage, and Christopher D. McDermott. 2026. Integrating Human Factors into Insider Threat Detection – A Systematic Review. *ACM Comput. Surv.* 58, 10, Article 260 (April 2026), 37 pages. <https://doi.org/10.1145/3798089>

1 Introduction

In the contemporary digital landscape, insider threats pose a critical security challenge for organisations. Insider threats originating from employees, contractors, or business partners with legitimate system access [24, 82, 102], are now regarded as the second most significant cybersecurity concern worldwide, according to **Chief Information Security Officers (CISOs)** [113], and have consistently been recognised as a key risk for both organisations and governments [106]. The financial impact of insider threats has escalated in recent years underscoring the importance of insider threat detection research. According to industry reported data (based on the Ponemon Institute’s 2025 Cost of Insider Risks Report) suggests that the average cost of insider-driven incidents in North America may reach USD 22.2 million annually.

Insider threats may be intentional, where individuals act maliciously to damage or exploit organisational assets, or unintentional, where errors such as mishandling information or neglecting security protocols inadvertently create vulnerabilities [45, 65]. Given insiders’ authorised access, both forms present heightened risks compared with external attackers. To address this insider threats problem organisations commonly deploy technical solutions such as encryption, hashing, and intelligent document systems [51], with most effort predominantly focused on technical dimensions [94]. However, human behaviour including errors, attitudes, and contextual influences, also plays a central role in shaping insider risk.

Human error contributes to a significant portion of security breaches, with 74% of breaches involving some form of human error. Notably, 43% of these cases are due to non-malicious actions, highlighting that many breaches result from accidental mistakes rather than intentional attacks [20, 112]. Despite the significant involvement of human factor elements in insider threats, studies to date addressing these aspects are limited. Additionally, current literature lacks a clear distinction between intentional and unintentional insider threats. A comprehensive discussion of unintentional insider threats is also necessary, as preventing them is particularly challenging due to their high unpredictability. Hence, there is a clear need for a comprehensive systematic literature review, that addresses all the aforementioned aspects. Therefore, this article aims to address these limitations in the current literature, including a lack of information regarding differentiating intentional and unintentional insider threats, as well as insufficient consideration of various aspects such as human factors compared to technical factors.

1.1 Contributions of the Study

This article presents a systematic review of insider threats that is comprehensive, concise, and readily accessible to researchers seeking new insights in the field. The review emphasises the integration of both technical and human dimensions of insider threats. The primary contributions of this study are summarised as follows.

- First, it provides a systematic synthesis of insider threat research, elucidating how existing studies differentiate between intentional (malicious) and unintentional (accidental) insider threats.
- Second, it develops a preliminary taxonomy of human factors, offering a comprehensive overview of behavioural and cognitive elements that contribute to insider threats.

- Third, it critically evaluates existing insider threat detection techniques, comparing their strengths, limitations, and capabilities in addressing both intentional and unintentional threats.
- Fourth, it investigates the extent to which **Machine Learning (ML)** approaches to insider threat detection incorporate human factors, identifying the datasets employed and emphasising the scarcity of integrated human–technical datasets.
- Finally, it outlines key implications for future research and practice, emphasising the need for new datasets and ML models that capture both technical and human dimensions to improve the prediction and mitigation of unintentional insider threats.

To this end, we formulate the following **Research Questions (RQs)**:

1.2 Research Questions

- RQ1 : How does existing research distinguish between intentional and unintentional insider threats?
- RQ2 : What human factors contribute to insider threats, and how have these been characterised in existing research?
- RQ3 : What are the strengths and limitations of current insider threat detection techniques?
- RQ4 : What datasets have been used in ML approaches for insider threat detection or prediction, and what extent do these approaches incorporate human factors?

1.3 Organisation of the Review

This systematic literature review paper is organised as follows: The background of the study and the related literature are discussed in Section 2, which also addresses the need for this survey article. Section 3 details the review methodology, including the paper selection process. Sections 4 to 6 are structured addressing the RQs. Section 4 defines insider threats and differentiates between intentional and unintentional insider threats. Section 5 discusses the human aspects of insider threats, including influential human factors contributing to these threats. In Section 6, insider threat detection techniques are discussed, with a focus on their technical aspects. Section 7 then examines the datasets used in these technical approaches. The discussion is presented in Section 8 along with the threats to the validity of the review. Finally, the research opportunities, and conclusions are discussed in Section 9.

2 Background and Related Literature Reviews

This section discusses the background of insider threats to an organisation and how various perspectives have been developed regarding insider threats. It offers a high-level understanding of insiders, insider threats, the impact on organisations, and how human error can lead to inadvertently becoming an insider.

2.1 Background

The definition of an insider threat relies on how insiders are defined in the study [57]. An insider could be current employees with legitimate access rights, former employees whose access may not have been fully revoked, contractors, and vendors operating with temporary or role-specific privileges and business partners with trusted access through organisation's digital space [24, 82, 102].

The literature presents multiple definitions of insider threats, reflecting the complexity of the concept. Pfleeger et al. [91] define an insider threat as “an action by an insider that threatens an organisation's data, processes, or resources in a disruptive or unwelcome manner”. Similarly, Greitzer and Frincke [41] describe it as “harmful acts that trusted insiders might carry out; for

example, something that causes harm to an organisation, or an unauthorised act that benefits the individual". Both definitions emphasise the potential for harm originating from individuals with legitimate access, though the latter further highlights the insider's capacity to act in self-interest as well as to damage organisational assets.

The aforementioned definitions primarily address insiders who act with malicious intent, thereby excluding individuals who inadvertently pose a threat, commonly referred to as unintentional insiders. Collins et al. define unintentional insider threats through two criteria: (1) the insider possesses legitimate access to the organisation's network and computing environment, and (2) the insider's actions lack malicious intent but nonetheless compromise organisational or information security [25]. In contrast, the **Computer Emergency Response Team (CERT)** Insider Threat Center adopts a broader definition, specifying four dimensions: (1) the identity of the insider, including current or former employees, contractors, or business partners; (2) the level of access, referring to individuals with authorised rights to organisational networks or data; (3) the nature of their actions or inactions, which are undertaken without intent to cause harm; and (4) the resulting impact, namely, harm that undermines the confidentiality, integrity, or availability of organisational information systems [19].

Both intentional and unintentional insider threats have a significant negative impact on organisational security, posing risks to tangible assets as well as to intangible assets such as reputation, intellectual property, customer relationship and organisational culture. For example the 2023 Insider Threat Report [114] indicates a 74% increase in insider threat incidents between 2022 and 2023, with the most critical organisational impacts including reputational damage, information loss, and operational disruption. Analysis of insider threat incidents highlight the prominent role of human factors in their occurrence. Similarly, the Verizon Data Breach Investigations Report 2023 [112] highlights the breaches involving human factors such as errors, misuse of privileges, use of stolen credentials, and social engineering. Despite their importance, existing literature reviews provide limited examination of these human factors. Consequently, the consideration of human factors is essential for understanding, detecting, and preventing insider threats within organisations.

2.2 Related Literature Reviews

Several literature reviews on insider threats focus on detection methods, particularly using **Deep Learning (DL)** and ML [10, 70]. Although some surveys examine the datasets used to detect and predict insider threats using technical approaches, only a few reviews consider the human factors that influence such threats. The review in [77] discusses the most recent on human factors, exploring social and behavioural issues in cybersecurity. While it does not specifically examine insider threats, it provides an interdisciplinary framework that aids understanding of human factors in cybersecurity. Since the human element is often seen as the weakest link in organisational security [100], focusing on insider threats can improve security and reduce risks.

The role of dark personality traits in influencing insider threat behaviour has been investigated using multiple personality models, including the Dark Triad and DSM-5 frameworks [48]. It discussed three models: the Dark Triad (3 traits), Hogan Development Scale (11 traits), and the DSM-5 Personality Inventory (25 narrow traits). Although the study offered guidance for organisations to assess dark traits, the conclusions were primarily based on expert opinions, potentially introducing bias limiting validity.

Lavanya and Sriram [70] discuss human factors in insider threat detection using DL techniques. While their study provides a useful categorisation of insider types and challenges, the analysis remains largely descriptive. Using **Gated Recurrent Units (GRU)**, they detected employee psychological behaviour, achieving 68% and 71% accuracy on Enron email and tweets datasets [67]. However, the methodological justification for choosing GRU over other architectures was limited,

and the reported performance suggests only moderate predictive capability. Although the paper identified **Long Short-Term Memory (LSTM)** and **Convolutional Neural Network (CNN)** as potentially more effective for insider threat detection citing their strengths in handling temporal dependencies and large datasets, this comparison was not empirically substantiated. Moreover, the study did not sufficiently consider unintentional insider threats or provide a deeper integration of human and psychological factors in the model design.

Extending this perspective, Yuan and Wu [130] discuss DL-based insider threat detection, noting the lack of real-world datasets, as the majority are synthetically generated through simulated attack sessions. They identified 11 commonly used datasets, categorised into five groups: masquerader-based, traitor-based, substituted masqueraders, identification/authentication-based, and miscellaneous malicious, all focusing on malicious insider threats. This heavy reliance on simulated data raises concerns about the ecological validity and generalisability of model performance in operational environments. No datasets for unintentional insider threats were found. The CERT dataset [22] is notable incorporating psychometric data like Big Five personality traits, although its limited scope and controlled conditions restrict the representativeness of real organisational behaviour. The study highlights DL models such as Deep **Feed Forward Neural Network (FFNN)**, **Recurrent Neural Network (RNN)**, CNN, and **Graph Neural Network (GNN)** and suggests future directions like few-shot learning, self-supervised learning, and deep survival analysis for early threat detection. However, while these directions are promising, the paper provides limited discussion on how such approaches could practically address data scarcity or enhance interpretability of two key challenges in insider threat research.

Building on other work by previous authors, Raut et al. [99] discuss insider threat detection, emphasising the integration of DL techniques with log-based anomaly detection. It provides an overview of available datasets, including the CERT dataset, and reviews key approaches like **Deep Belief Networks (DBN)**, Autoencoders, RNN, and CNN. While DBN demonstrated reasonable performance, it struggled with time-series data and false alarms. Autoencoders reduced false alarms, and RNNs were most effective for temporal log data, with CNN-RNN combinations proving powerful. However, the comparative evaluation across these models was largely descriptive, offering limited quantitative analysis or benchmarking, which restricts the strength of the conclusions. The authors suggest integrating the **Elasticsearch-Logstash-Kibana (ELK)** toolset with DL models for industrial applications, yet the feasibility and scalability of such integration were not empirically examined. Moreover, the study overlooks the human element and unintentional insider threats which are critical for developing comprehensive detection frameworks that account for behavioural and contextual nuances.

In addition to the surveys previously discussed, other studies on insider threat detection [2, 66, 86] focus on ML-based approaches. In [2], the authors present a taxonomy of insider types, access levels, motivations, and attack methods, reviewing related works on ML techniques, datasets, and evaluation metrics. They also highlight challenges and recommend incorporating the human element in threat detection. The study in [86] discusses ML techniques, while [66] explores both ML and DL techniques. Most studies mainly focus on malicious insiders, neglecting unintentional threats and the human factor. While [53] provided a comprehensive review of insider threats, encompassing both intentional and unintentional manifestation, it provided limited coverage of recent studies and placed less emphasis on the human element. It did not provide updated datasets. The review included definitions, taxonomies, and detection strategies, but many datasets were outdated. This article served as a benchmark for our study, focusing on technical papers from 2019 to 2025 to update knowledge on insider threats.

Insider threat attacks alongside corresponding prevention and detection techniques have been examined by [59] presenting a taxonomy that classifies insider threats by type, access level, and

target security objective. The study emphasises that negligent insiders are more difficult to address than malicious ones and suggests that enhancing staff training and education on fundamental security practices can help mitigate accidental insider threats. Consideration of human factors is also identified as a key future direction and challenge, underscoring their necessity in insider threat prevention.

The systematic review in [63] analyses 101 studies on **User Behavioural-Based Insider Threat Detection (UBITD)**, offering a comprehensive synthesis of ML/DL techniques, preprocessing, evaluation methods, and datasets. Strengths include structured methodology, broad dataset coverage (CERT, TWOS, RTU), and clear research gaps. Findings show ML (**Support Vector Machines (SVMs)**, **Decision Trees (DT)**) and DL (LSTM, autoencoders) perform well for sequential/high-dimensional data but rely heavily on supervised learning with labelled datasets, limiting adaptability. Key limitations are scarce real-world datasets, imbalance issues, and vulnerability to adversarial insiders. The review calls for advanced preprocessing, imbalance handling (SMOTE, cost-sensitive methods), explainability, and hybrid ensembles to reduce false positives. Reported results show promising accuracy/recall, but inconsistent metrics hinder comparisons. Future work should pursue synthetic data generation, semi/unsupervised methods, and domain-specific datasets (IoT, cloud) for scalability and real world use.

By reviewing 99 papers on healthcare data breach mitigation, the study in [84] highlights strengths such as the use of advanced technologies including Blockchain, AI, and encryption for enhancing security resilience. Limitations include lack of practical implementation, weak regulatory alignment, and poor employee training, leaving human factors vulnerable. Most works rely on encryption (81%) and hash functions (56%), though many remain theoretical without real world validation. Dataset use is limited, with few studies on real healthcare data. Future directions include addressing IoT/cloud security, preparing for quantum threats, and building deployable, regulation compliant architectures with strong verification. Aligning with this view, Alketbi et al. [3] shows that **Explainable AI (XAI)** improves malicious insider threat detection by enhancing transparency, trust, and accountability. Strengths include structured insights into ML/DL/hybrid methods, benchmarking datasets (CERT, CICIDS2017, UNSW-NB15), and effective use of **SHapley Additive exPlanations (SHAP)**, **Local Interpretable Model-agnostic Explanations (LIME)**, and RuleFit for interpretability. Limitations involve high false positives, contextual blindness, imbalance, privacy concerns, and lack of standardised metrics. Models often achieve high accuracy/F1 on benchmarks, though results vary across domains. Enhancements include multimodal/psychometric data integration, privacy preserving and federated learning, human-in-the-loop interpretability, and standardised explainability benchmarks. Overall, XAI-enabled systems achieve strong detection utility, but long-term effectiveness depends on data quality and balancing accuracy with interpretability.

In contrast to the previously discussed studies, insider threats in the UK financial services sector have been examined by Whitelaw et al. [123], with a more focus on policies and procedures rather than technical detection measures. The study noted the widespread use of insider risk programs and technical controls, yet found that significant and costly risks persist. Strengths include blending practitioner perspectives with academic findings across five themes: behavioural indicators, security behaviour, technical controls, strategies, and regulation. Limitations are few FS-specific studies, reliance on generic literature, ethical/privacy concerns in monitoring, and limited evidence of impact. Suggested improvements include psychosocial/organisational factor integration, cross-functional collaboration (IT, HR, compliance), tailored risk segmentation, and emerging tools (UEBA, sentiment analysis, deception). Technical methods (ML, UEBA, Bayesian models) show strong accuracy in controlled studies, but real-world impact is mixed due to cost and maturity. Datasets rely on case studies, incident reports, and synthetic benchmarks (CERT), with

no large-scale FS-specific datasets, limiting validation. The study stresses people-centric, adaptive, evidence-based approaches to strengthen resilience.

The advances in malicious insider threat detection using ML have been summarised by the review in [6]. Traditional methods (rule-based, anomaly detection, supervised/unsupervised) laid the foundation, while DL, NLP, and graph-based approaches now achieve superior accuracy/adaptability. Strengths include advanced models (LSTMs, autoencoders, transformers) capturing temporal/contextual patterns, improving detection and reducing false negatives on benchmarks (CERT, RUU, TWOS, synthetic). Limitations include data scarcity, reliance on synthetic datasets, false positives, interpretability issues, and adversarial evasion. DL/hybrid approaches outperform classical ML with high accuracy/recall/AUC (>0.90) on CERT, though real-world validation is scarce. Future work should focus on XAI integration, privacy-preserving/federated learning, multi-source data fusion (logs, text, behaviour), and realistic, shareable datasets. Overall, ML progress is strong, but bridging the gap between synthetic and real-world environments remains essential.

2.2.1 Summary of Related Literature Review. A considerable body of literature has examined insider threat detection, with reviews emphasising a range of approaches spanning ML, DL, behavioural analysis, and human factors. Surveys such as those by Lavanya and Sriram [70], Yuan and Wu [130], and Raut et al. [99] highlight the growing sophistication of DL models GRUs, LSTMs, CNNs, RNNs, and hybrids, yet consistently identify limitations relating to data scarcity, over reliance on synthetic benchmarks (e.g., CERT), and insufficient consideration of unintentional insider threats. Complementary reviews have focused on the human dimension, exploring psychosocial and behavioural issues [77], dark personality traits [48], and sector specific insights such as healthcare [84] and financial services [123]. These studies reinforce the importance of human centric perspectives but are often constrained by reliance on expert opinion, theoretical frameworks, or limited empirical validation. Other systematic surveys [2, 63, 86] provide structured taxonomies, comprehensive coverage of ML/DL techniques, and critical evaluation of preprocessing and evaluation methods, but similarly acknowledge shortcomings such as imbalance issues, lack of real world datasets, and neglect of the human element. Emerging directions, including XAI [3], hybrid ensembles, and integration of organisational and psychosocial factors, are increasingly recognised as vital to bridging technical accuracy with interpretability, trust, and operational impact.

As summarised and compared in Table 1, existing reviews provide valuable insights into related studies. However, unintentional insider threats and the human element remain less well understood compared to technical factors, and this knowledge is fragmented across multiple sources, making it difficult to access in a comprehensive manner. Moreover, many recent studies have yet to be systematically reviewed, and no single work offers an integrated perspective that brings together both technical approaches and human-centred considerations. This narrow focus limits the field's ability to understand insider threats as multidimensional phenomena shaped by behavioural, organisational, and technical influences. To address this gap, the present review consolidates these fragmented strands of research and provides an integrated perspective that considers both intentional and unintentional insider threats, situating human factors alongside technical and organisational elements within a single, comprehensive synthesis. Such a review is essential to advance understanding, support holistic detection strategies, and guide future research in building more resilient insider threat mitigation frameworks.

3 Methodology

In this review, the **Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA)** method [88] was used to report and analyse relevant research studies. The databases

Table 1. Summary of Related Reviews

Paper	Focus	Insider Threats											
		Intentional			Unintentional			Detection Techniques					
		Taxon-omy	Challe-nges	Human Factors	Taxon-omy	Challe-nges	Human Factors	DL	ML	Other AI	Statis-tical	Manu-al	Hybrid
Homoloak et al. (2019) [53]	Taxonomies, analysis, modelling, and countermeasures	✓	✓	✓	✓	✓	✓						
Kim et al. (2019) [66]	Insider threat detection	✓											
Oladimeji et al. (2019) [86]	Insider threat detection techniques								✓			✓	
Al-Mhiqani et al. (2020) [2]	Detection,techniques, datasets, challenges, and recommendations	✓	✓						✓	✓		✓	
Raut et al. (2020) [99]	DL-based insider threat detection							✓					
Subhani et al. (2021) [115]	Insider threat detection in organisations		✓						✓				
Yuan and Wu (2021) [130]	DL-based insider threat detection		✓					✓					
Harms et al. (2022) [48]	Linking dark traits to insider threat prediction			✓									
Lavanya and Sriram (2022) [70]	DL-based insider threat detection							✓					
Marbut and Harms (2023) [78]	Perspective on personality and insider threats			✓									
Alzaabi et al. (2024)[6]	Malicious insider threat detection		✓					✓	✓				✓
Inayat et al. (2024) [59]	Insider threat attacks, prevention & detection techniques	✓	✓					✓	✓				
Whitelaw et al. (2024) [123]	Insider threats in UK financial services		✓	✓		✓	✓						
Alketbi et al. (2025) [3]	Explainable AI (XAI) for insider threat detection		✓	✓				✓	✓				✓
Kamatchi et al. (2025) [63]	User behavioural-based insider threat detection		✓	✓					✓	✓			
This work	Analysis of insider threats incorporating human and technical perspectives	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

used, along with the inclusion and exclusion criteria, were also specified. The study selection, data extraction, and synthesis were carried out to answer the formulated RQs. In addition, threats to the validity of this systematic review of the literature and mechanisms to reduce bias and other factors that could have influenced the results of this study were identified.

3.1 Search Strategy

This review used six top academic databases including ACM Digital Library,¹ IEEEExplore Digital Library,² Web of Science,³ Science Direct,⁴ Springer,⁵ and Scopus.⁶ Google Scholar was used to identify additional relevant papers during the snowballing process, but was not used in the initial search, as it aggregates content from multiple sources, many of which overlap with results from other academic databases.

A designated search string was used to browse through these academic databases and initially, studies published from 2014 to 2025 are considered. The search string employed was: (“insider threat”

¹<https://dl.acm.org/>
²<https://ieeexplore.ieee.org/>
³<https://www.webofscience.com/>
⁴<https://www.sciencedirect.com/>
⁵<https://link.springer.com/>
⁶<https://www.scopus.com/>

Table 2. Search Results Distribution with Duplicates

Repository	Number of Papers	Duplicates
Springer	859	0
Scopus	174	83
Web of Science	62	33
ACM	49	1
IEEE	241	2
Science Direct	74	0
Total	1459	119

OR “accidental insider*” OR “unintentional insider*”) AND (“influential factors” OR “behaviour*” OR “human factors”). The search results and the number of duplicated records are illustrated in Table 2.

3.2 Study Selection, Data Extraction, and Synthesis

Since this review focuses on two main aspects, two survey strategies were followed when selecting studies. Phase 1 of this review concentrates on the human aspects of insider threats, with a focus on distinguishing between intentional and unintentional insider threats, which are under-explored in this domain. Although the authors in [53] discussed some human factors and unintentional insider threats, a more comprehensive analysis is necessary. Therefore, under this phase, the review covers human-related aspects of insider threats, including both intentional and unintentional threats, from 2014 to October 2025, due to the scarcity of studies in this area. Phase 2 focuses on the technical aspects of insider threats, such as detection techniques, ML and AI approaches and relevant datasets. This phase is detailed in Sections 6 and 7. To establish the inclusion and exclusion criteria for this phase, a benchmark review paper [53] was consulted, which provided a broad overview of insider threat research up to 2019. Accordingly, the second phase considered studies published between 2019 and 2025, as most earlier work had already been addressed in the benchmark review. The objective of this phase is to examine the technical perspectives on insider threats within this time frame.

To ensure a focused and relevant selection of studies for this review, specific inclusion and exclusion criteria were applied.

- Studies with a primary focus on insider threats, particularly those impacting organisational security
- Papers addressing influential factors related to insider threats, such as individuals’ information security practices, and those exploring insider threat detection and mitigation techniques
- Studies classified as insider threats within specific domains, such as cloud computing and IoT, were excluded
- Papers were required to be written in English and have a minimum length of three pages

We required publication in peer-reviewed, indexed venues. Rankings (impact factor, CORE rank, publisher reputation) were also considered as contextual indicators during screening.

Initially, 1459 records were found from the search, and 119 records were excluded due to duplication. The remaining 1340 records were carefully screened based on the aforementioned eligibility criteria, and 158 were selected, excluding 1182 records. These 158 papers included 104 technical papers and 54 human and unintentional insider threat-related papers. Then, those 158 papers were reviewed, and among them, 26 technical papers and 21 human aspects related papers were excluded due to insufficient analysis. After that process, the remaining technical papers numbered 78 and human aspects related papers numbered 33. Additionally, during this review, we identified and included related papers through snowballing to enhance the clarity of the discussion. Through

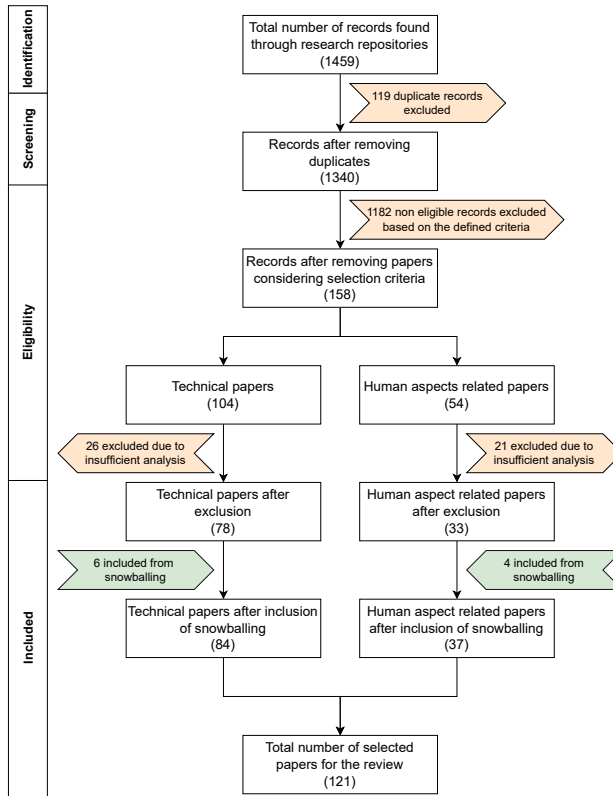


Fig. 1. PRISMA method:paper selection.

snowballing, 6 additional technical related papers and 4 human aspects related papers were included. Finally, after all these steps of PRISMA, this study reviewed a total of 121 papers. Figure 1 shows a summary of the paper selection method for this systematic review.

4 Differentiating Intentional and Unintentional Insider Threats

This section discusses the definition of insider threats while differentiating intentional and unintentional insider threats. It also discusses various insider threat taxonomies proposed in the literature.

4.1 Defining Insider Threats

After a thorough investigation of different definitions, insider threats can be defined as individuals within an organisation's security perimeter, such as employees, ex-employees, contractors, and auditors, who have special privileges or access to the organisation's technological infrastructure, who engage in information security malpractice [26]. Typically, insider threats are categorised as malicious or non-malicious based on whether the actions carried out by the individual are intentional or unintentional.

4.2 Insider Threat Taxonomies

Various taxonomies of insider threats have been proposed. Some taxonomies are built considering both intentional and unintentional insider threats, while others are created based on a single

type of insider threats. For instance, the taxonomy discussed in [93] addressed both categories of insider threat. In this study, insider threats have been grouped as accidental, negligent, mischievous, and malicious behaviours through thematic analysis of the literature. These four categories are evaluated based on six aspects: intention, motivation, use of privileges, skill in using technology, causing factors, and awareness or unawareness of organisational rules. Another taxonomy in [110] considered both aspects of insider threats. In this study insider threats are divided into three categories as (1) Individuals with an intention to carry out malicious actions, (2) Individuals motivated by personal gain, (3) Individuals who inadvertently or unknowingly engage in harmful behaviour.

Recent studies have also increasingly focused on developing comprehensive taxonomies that combines multiple aspects of insider threats. Among them, the study [1] has classified seven aspects : insider access, insider motivation, insider attack indicators, insider types and actions, insider profile, insider methods, and insider threat detection techniques. Similarly, in [79], insider threats have been explained based on four categories: individual, action, asset, and consequence, further distinguishes whether the threat is intentional or unintentional within each category.

The taxonomy by CERT [42], specifically addresses unintentional insider threats. There are four types of unintentional insider threats: malicious code, information disclosure, improper or accidental disposal of physical records, and portable equipment no longer in possession. Furthermore, another study [119] has suggests that unintentional insider threats can be classified into two categories: well-meaning and negligent. However, there is a notable conflict exists between [93] and [119] because, in [93], the intention of negligent insiders is considered intentional, whereas [119] considers negligent actions as unintentional. Additionally, the study [119] has mentioned four categories of employees capable of causing data spillage: (1) underminers - those who do not respect organisational security policies to make their lives easier, (2) overambitious - those who avoid following security measures, (3) socially engineered - those who can be victims of social engineering, and (4) data leakers - those who make data available in public, perhaps due to accidents or through public posts.

However, only a limited number of taxonomies have been developed to distinguish unintentional insider threats. Most taxonomies in the literature focus on broader insider threat domain or on intentional/malicious insider threats in isolation entire insider threat domain or isolating intentional/malicious insider threats. In the literature, several malicious insider threat taxonomies have been built based on insider misusers, insider attack types, insider threat knowledge, categorising insider threats by network monitoring, and types of motivations [53]. Hence, the literature highlights a clear need to further investigate unintentional insider threats.

5 Human Aspects of Insider Threats

This section discusses the definition of human factors and the influential human factors for insider threats, while introducing a novel taxonomy.

5.1 Defining Human Factors

Human factors research has long been recognised as essential for ensuring safe and effective performance in the workplace across diverse domains such as aviation, healthcare, and transport [14, 31, 38, 122]. More specifically it has been applied in computing, and, more broadly, within the context of Industry 4.0 [62, 92].

Human factors constitute an interdisciplinary research area that integrates disciplines such as psychology, engineering, physiology, and ergonomics, which examines human abilities, limitations, and characteristics that are important for human performance, safety, and design. This approach can also be used to design tools, machines, systems, tasks, jobs, and environments that are safe, comfortable, and effective for people to use. The National Aeronautics and Space Administration (NASA) has

defined human factors as a broad term encompassing research in human performance, technology design, and human–computer interaction. For this review study, we define human factors associated with insider threats as elements related to individual, organisational, situational, task-related, and motivational factors that contribute to the risk of both intentional and unintentional insider threats. The categorisation was guided by human factor classifications from various research domains, with particular influence from the ontology-based model **Sociotechnical and Organisational Factors for Insider Threat (SOFIT)**, developed by Greitzer et al. [40]. Furthermore, the new categorisation was subsequently peer-reviewed by domain experts.

5.2 Influential Human Factors for Insider Threats

The literature discusses several influential human factors associated with insider threats, employing various categorisation methods. This section will examine human factors influencing both intentional and unintentional insider threats, grouped into individual, organisational, and situational and task-related factors. Additionally, we introduce a separate category of motivational factors, which are specifically related to intentional insider threats.

5.2.1 Individual Factors. Greitzer et al. [43] conducted to identify factors influencing unintentional insider threats related to social engineering, identified three broad categories of human factors: cognitive limitations, biases, and psychological traits. When developing the SOFIT ontology, the focus was placed on individual factors such as behaviours, attitudes, personal issues, sociocultural or ideological factors, and biographical details [40]. A mixed methods study examining the behavioural indicators that occur before an insider act happens identified behavioural changes, workplace behaviours, and personality traits as key factors [16]. The study further suggested that intentional insider threats are often associated with deliberate malicious actions and personality traits, while unintentional threats may stem from factors like financial stress and substance abuse. However, the study only measured behavioural intention rather than actual behaviour, introducing subjectivity. Moreover, individual differences under behavioural factors were discussed in the study [108]. This study explored individual differences, such as personality traits (e.g., conscientiousness, extraversion, openness), and motivational factors influencing unintentional insider threats. To support this, the Cybersecurity Self-Efficacy and Cybersecurity Knowledge (CSEC) questionnaire was developed and helped organisations tailor cybersecurity training. The study revealed that individuals with high levels of fear and anxiety, coupled with low moral motivation, were more likely to engage in risky behaviours, contributing to unintentional insider threats. However, the high subjectivity of self-reported data was a limitation.

Using a grounded theory approach, [39] explored human factors in cybersecurity and identified cultural, technical, and individual contributors to insider threats, resulting in four main themes: awareness, familiarity, ignorance, and lack of training. Among these, employees' awareness was highlighted as a crucial factor in preventing unintentional insider threats. The study also identified that some employees committed cybersecurity infractions either unwittingly or due to factors such as attention deficits, lack of caring, complacency, and ethical dilemmas occur when a person faces a choice between conflicting moral principles or obligations. Similarly, decision-making complexity, and cognitive lapses (e.g., lack of awareness, reduced attention) were highlighted as significant contributors. However, this study's relatively small sample size ($n=15$), limited the generalisability of its findings.

Apart from the studies discussed above, several other direct and indirect studies have been conducted to identify influential human factors contributing to insider threats. Among them, Hills and Anjali [52] identified behavioural changes, cognitive biases, substance abuse, personal problems, and even cultural and social media influences as relevant factor. A study on human

factors in information leakage, which can be indirectly linked to insider threats due to information disclosure, identified personal greed, jealousy, and disgruntlement as causes of intentional leakage, while human error, negligence, over-enthusiasm, and lack of awareness were recognised as factors leading to unintentional leakage [124]. Studies such as [4, 21, 50, 65, 81, 83, 85, 87, 118, 128] have also discussed individual factors that fall under human factors, which can influence insider threats.

Overall, these studies underscore the complex interplay of behavioural, cognitive, and emotional factors in insider threat incidents. Yet, limitations such as reliance on behavioural intentions, small sample sizes, and self-reported data suggest that further research is needed to solidify these findings and enhance their practical application in cybersecurity training and policy development.

5.2.2 Motivational Factors. Motivational factors play a vital role in shaping insider threats, particularly in cases of intentional breaches [107]. While these factors could arguably be treated as a subset of individual factors, there is clear justification for addressing them separately in the context of insider threats. In most insider threat classifications, the distinction between intentional and unintentional insider threats is primarily determined by the actor's underlying intention or motivation. For this reason, it is essential to place particular emphasis on motivational factors when examining the dynamics of insider threats.

Personal gain, financial reward, addictions, and disgruntlement have been identified as key motivators driving individuals to commit malicious acts against their organisations [29, 40, 45]. For instance, financial reward is a significant motivator, with employees engaging in insider threats for personal enrichment, often through actions such as embezzlement or data theft [45]. Addictions, whether to substances or gambling, can also push individuals toward insider threats as they seek to fund their habits [29]. Disgruntlement, arising from dissatisfaction with the organisation poor management practices or job dissatisfaction, is another powerful motivator that can lead employees to engage in retaliatory actions against their employer [40]. These individuals may rationalise their actions as justified revenge or as a means to address perceived injustices within the workplace. Collectively, these motivational factors highlight the complex interplay between personal circumstances and organisational environments, illustrating the need for comprehensive insider threat programs that address both technical and human elements to mitigate these risks.

5.2.3 Situational and Task Related Factors. Task complexity, time pressure, job pressure, and time planning emerge as significant contributors to both intentional and unintentional insider threats [29, 40, 45].

Task complexity is often linked to the cognitive load required to complete specific duties, with more complex tasks increasing the likelihood of mistakes and unintentional security breaches. For instance, excessive task complexity may overwhelm employees, reducing their attention to security protocols and raising the probability of human error, as seen in incidents where large amounts of data are mishandled or improperly stored [45]. When tasks are too complex, employees might inadvertently neglect critical security measures, which can lead to vulnerabilities that are exploited, either accidentally or intentionally. Time pressure represents another crucial factor that exacerbates the risk of insider threats. Under tight deadlines, employees may prioritise task completion over compliance with security protocols, leading to shortcuts that bypass essential security measures [29]. The urgency to meet deadlines can further cause employees to overlook potential security risks, thereby increasing the likelihood of both intentional and unintentional insider incidents. For instance, employees might resort to insecure practices, such as sharing passwords or skipping two-factor authentication, when they feel pressured to complete tasks swiftly. Job pressure and workload, while closely related to time pressure, encompass broader issues such as employee stress, burnout, and overall mental health [29]. Excessive workloads can contribute to lapses in judgement, where individuals may unintentionally compromise security due to fatigue or stress

[40]. Moreover Job pressure can also motivate intentional insider threats, particularly in situations where employees feel undervalued or overwhelmed by their work, potentially leading to malicious actions as a form of retaliation.

Time planning and poor task scheduling are additional situational factors that influence the risk of insider threats. Inefficient time management or inadequate planning can lead to situations where employees are forced to rush through tasks or work outside of standard procedures [40]. Such inadequate planning often produces uncoordinated workflows, leading employees to make hasty decisions that compromise security protocols. The interplay between these situational and task-related factors underscores the importance of addressing organisational practices that contribute to insider threats. Addressing these factors through improved organisational practices can significantly reduce the risk of both intentional and unintentional insider threats.

5.2.4 Organisational Factors. Various aspects of a company's structure and practices are encompassed under organisational factors. The literature identifies several key factors, including organisational culture, policies, workplace dynamics, job pressure, and leadership. Elmrabit et al. [29] developed a malicious insider threat prediction framework that incorporates organisational factors such as job dissatisfaction, lack of security policies, and poor management practices. Employing **Bayesian Network (BN)** statistical methods to predict insider threats, organisational factors were found to be crucial in both intentional and unintentional insider threats. Moreover, work-related stress and poor policies were found to contribute to unintentional threats. In addition, organisational factors such as inadequate training, poor communication, and ineffective policies have also been highlighted as important in the literature [43, 65].

In examination of contributing factors, observables, and mitigation strategies for unintentional insider threats, Greitzer et al. [42] collected cases from public sources and the CERT insider threat database. Using a feature model, organisational factors such as inadequate procedures, poor communication, job pressure, and task difficulty were identified as critical contributors. Leadership and organisational culture were also highlighted as key influences on insider threat risks. Themes like awareness, selfishness, and care are being linked to organisational culture, impacting both intentional and unintentional insider threats [43]. The study further underscored the importance of employee readiness, stress management, fatigue reduction, and skill development in mitigating insider threats.

5.3 Human Factors of Insider Threats - The Taxonomy

Although human factors have been widely examined in the literature, a clear distinction between the factors influencing intentional and unintentional insider threats remains underdeveloped. This review addresses this gap by categorising influential factors according to insider threat motivation, distinguishing between intentional and unintentional cases, a preliminary taxonomy was developed.

The existing SOFIT taxonomy primarily classifies insider incidents by observable behaviours and technical-organisational conditions but gives limited emphasis to the cognitive mechanisms underlying unintentional errors. Furthermore, their categorisation does not differentiate between contributing factors for intentional and unintentional insider threats. A clearer classification is therefore required, one that explicitly considers these two dimensions.

As an initial step in taxonomy development, contributing factors for intentional and unintentional insider threats were identified separately. Following the SOFIT framework, these factors were subsequently grouped into categories. Figure 4 illustrates the human factors contributing to intentional insider threats, while Figure 5 illustrates those contributing to unintentional insider threats, as identified in the literature. These factors are categorised into three main groups according

to the SOFIT categorisation technique, which aligns with the **Sociotechnical Model (STM)**. The categories include individual (human) factors, organisational factors, and situational and contextual factors.

Consequently, the identified factors were considered alongside their categories, and a new categorisation was introduced based on hierarchical models, which are essential for systematically analysing human factors and their interactions within complex systems [90]. These models provide a structured approach to understanding how individual, team, organisational, and environmental factors contribute to human behaviour and performance, making them valuable tools in various fields, including safety, ergonomics, and organisational development. This approach was adopted in the categorisation of human factors in the present study and the developed taxonomy was validated by subject experts.

Individual factors encompass personal characteristics such as cognitive limitations (e.g., lack of knowledge, memory failures), behavioural and emotional states (e.g., stress, prevention focus), and biographical or sociocultural attributes (e.g., age, gender, and cultural influences). These factors represent internal traits and personal circumstances that may influence risk-taking behaviour or lapses in awareness. Organisational factors reflect workplace-driven elements, including the organisation's culture, policies, and leadership, such as inconsistent security policies and poor communication. These factors shape the environment in ways that can either contribute to or mitigate insider threats, depending on how supportive or structured the organisation's practices are. Finally, situational and contextual factors relate to external and environmental conditions impacting job performance and stress, such as task complexity, excessive workload, and inadequate training or resources. This categorisation helps pinpoint specific intervention areas and allows for tailored solutions to address vulnerabilities across individual, organisational, and situational levels, thereby providing a holistic understanding of insider threat contributors.

A close comparison of all these factors highlights overlapping influences that affect intentional and unintentional insider threats. These factors include stress, substance abuse, organisational issues such as poor management practices and inadequate training, cognitive limitations such as misjudgement and faulty reasoning, and emotional states such as anxiety and fear. Considering these factors and their contributions to both intentional and unintentional insider threats, a taxonomy of human factors for insider threats is created as illustrated in Figure 2.

6 Insider Threat Detection

This section provides a detailed in-depth exploration of the various techniques used to detect insider threats, as well as the challenges posed by insider threat detection.

6.1 Insider Threat Detection Techniques

Based on search results from various repositories, recent studies conducted after 2019 have increasingly focused on insider threat detection. A significant portion of these studies emphasises the integration of AI, ML, and DL techniques to enhance detection capabilities. Apart from them statistical based methods, manual methods, and heuristic based methods were also used. This section discusses these insider threat detection techniques, considering various approaches.

6.1.1 DL, ML, and Other AI-Based Detection Approaches. Based on the search results, it is observed that recent studies conducted after 2019 have placed greater emphasis on insider threat detection. Most of these studies have focused on integrating AI techniques to detect insider threats. Unlike traditional methods which rely on predefined patterns, AI-based methods can analyse a large volume of data, learning from both historical and real time data. These approaches allow for more proactive detection of both malicious insiders and unintentional breaches.

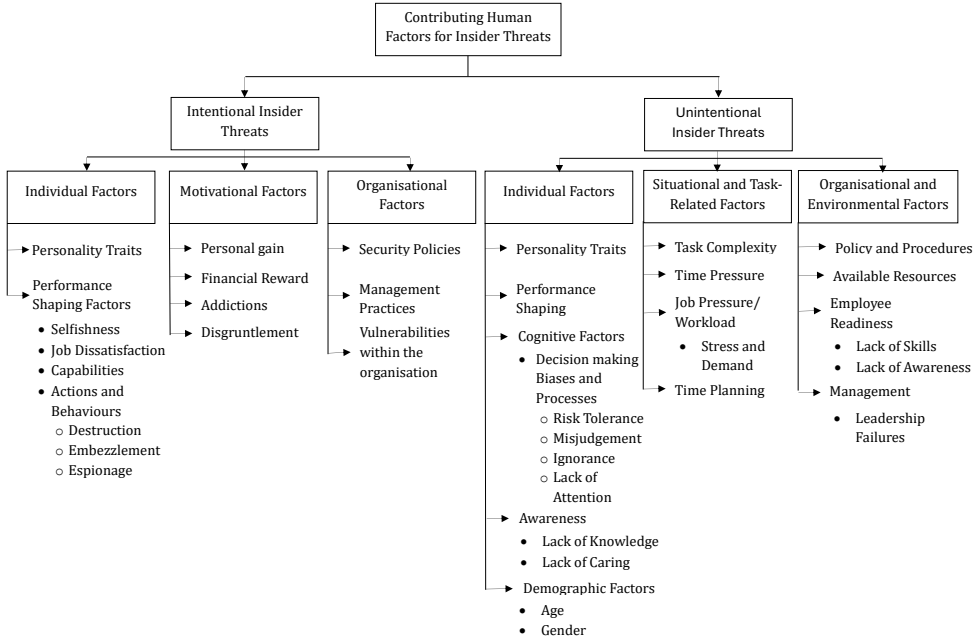


Fig. 2. Taxonomy of human factors of insider threats.

The ML lifecycle includes data extraction, preprocessing, feature selection, model training, evaluation, and deployment [7]. ML methods are categorised as supervised, unsupervised, semi-supervised, reinforcement, and DL. Supervised learning uses labelled data for classification and regression with algorithms like **Naive Bayes (NB)**, **Logistic Regression (LR)**, SVM, DT, **Random Forest (RF)**, **eXtreme Gradient Boosting (XGB)** and **k-Nearest Neighbors (kNN)**. Unsupervised learning identifies hidden patterns in unlabelled data using techniques like K-means clustering, **Principal Component Analysis (PCA)**, and autoencoders. Semi-supervised learning combines labelled and unlabelled data, while reinforcement learning adjusts model parameters through environmental feedback [7]. DL, characterised by multilayered complex algorithm analysis, includes models like CNN, LSTM, RNN, **Generative Adversarial Networks (GAN)**, and **Multilayer Perceptrons (MLP)**, offering high accuracy in detecting subtle, evolving threat patterns [130].

Feature engineering is important for AI-model building to detect insider threats. A comprehensive feature engineering solution was introduced in [54], based on users' daily activities and incorporating both manual and automated feature extraction. Sequential activity features were automatically extracted using an LSTM auto-encoder, which, when compared to manual methods, improved the F1-Score by 0.56% in identifying hidden patterns. The proposed model, ResHybnet, a hybrid network combining GNN and CNN elements, achieved 92.62% accuracy and an F1-Score of 92.19 on the CERT 4.2 dataset, surpassing classifiers like SVM, LR, RF, GNB, CNN, and GCN. Key factors behind ResHybnet's performance included: GNN's ability to detect topological connections in organisational graphs, its capacity to retain node attributes via residual links, and CNN's capability to recognise patterns in non-topological data. Although effective, the study noted limitations, such as reliance on the CERT 4.2 dataset and unaddressed data imbalance, suggesting that online

learning could help mitigate these issues. Similarly, [5] focused on reducing false positives and negatives by integrating feature extraction, sentiment analysis, and Gray encoding. Their framework achieved a false positive rate of 0.29%, a false negative rate of 2.47%, and an AUC of 97%. However, practical scalability and adaptability to diverse organisations remain unexplored, with limitations tied to the CERT r4.2 dataset and the complexity of Gray encoding, which may hinder real-time detection. Future work should test this framework in varied environments to enhance real-world applicability.

Randive et al. [96] proposed a **Wavelet Convolutional Neural Network (WCNN)** that transforms user access logs into image-based representations, achieving high performance on the CERT 4.2 dataset (97.19% accuracy, 99% recall, AUC 97.3%). Similarly, Zheng et al. [132] introduced a heterogeneous graph embedding approach that maps user behaviour sequences into high-dimensional feature spaces, outperforming several baselines on the CERT r5.2 dataset (AUC 0.98, F1-score 0.89). While both methods show promising results, further research is needed to validate their generalisability beyond specific datasets. The authors in [95] proposed a ML and DL approach to create persona-specific online traffic profiles, enabling persona identification and anomaly detection without extra software. The method segments network traffic from insider devices using models like FFNN, CNNs, LSTM, DT, RF, and ensemble classifiers. Tested on real-world data, it achieved 97.96% accuracy in distinguishing user from device generated data, 98.89% in user profiling, and 98.91% in cyber persona identification, with an AUC of 99.95% for detecting insider threats. Future improvements could enhance model performance and incorporate website context for embedding vectors. The paper [129] presents the **Deep Adversarial Insider Threat Detection (DAITD)** framework, which uses GANs to create synthetic samples of anomalous behaviour, addressing the challenge of class imbalance. Traditional re-sampling techniques like SMOTE often lead to over-fitting due to lack of sample diversity. DAITD includes three components: user behaviour representation (using an LSTM-autoencoder), data augmentation with GAN, and classification. GAN generates synthetic anomalous samples to balance the dataset, while classifiers like DT and XGB are used for threat detection. Testing on the CERT dataset, DAITD achieved an average F1-Score of 0.9837 (XGB+GAN) and 0.9755 (DT+GAN). Further testing on more scenarios and datasets is recommended to validate its robustness.

Authors in [76] present a method for detecting insider behaviour by analysing system logs with a specialised RNN. System logs are treated as natural language sequences to identify patterns, creating user-specific log sequences to distinguish normal from abnormal behaviour. Tested on the CERT dataset, the model achieved 93% accuracy in detecting insider threats. Future improvements could include continuous learning strategies, such as online learning, to better handle diverse and emerging user behaviours, potentially reducing false positives and false negatives. Another study [64] presents a user behaviour analysis system covering data processing, behaviour modeling, and results analysis. It introduces a data adjustment strategy to address data imbalance, including suspicious data selection and combination sampling, enabling effective employee behaviour analysis for hidden threat detection. The **Ethnic Randomised Practical Swarm Optimisation (ERPSO)** algorithm optimises the model, tested on the CERT 4.2 dataset with user activities like file access and logon events. The ERPSO-XGBoost model outperformed state-of-the-art methods, offering insights for enterprise management and dynamic strategy adjustments to prevent insider threats. Future work could include diverse datasets to improve generalisation.

An insider threat detection method using GRUs and multi-autoencoders to address imbalanced datasets by capturing temporal correlations in user behaviour was proposed by [80]. The model outperforms traditional methods like Isolation Forests and LSTM autoencoders, achieving better efficiency and accuracy with multi-head attention. However, it lacks exploration of scalability and real-world applicability, and unsupervised learning may limit adaptability to dynamic threats.

Similarly He et al. [49] introduced LSTM and Bi-LSTM models with attention, achieving a 3.1% to 6.3% improvement over previous methods. Yet, reliance on the CMU-CERT dataset limits generalisability, and broader testing across multiple datasets is recommended. Pal et al. [89] also employ stacked-LSTM and stacked-GRU models with attention to capture sequential and contextual user behaviour features, addressing prior limitations such as simple feature reliance and data imbalance. While detection accuracy improves, the study could further explore computational complexity and real-time scalability, as well as attention interpretability for actionable insights.

In the context of classical ML approaches, Li et al. [72] proposed a model based on timestamps and user identities for detecting insider threats in university web clusters outperforming traditional methods in precision and F1 scores. However, its accuracy is limited by a small sample size and unrepresented events, and further refinement is needed to reduce false positives and validate the model in real-world scenarios. Dixit et al. [27] evaluated KNN on the CERT dataset, achieving 93.5% accuracy, but reliance on a single dataset and lack of comparison with methods like RF restrict generalisability. Similarly, Janjua et al. [60] employed supervised learning using Adaboost on the TWOS email logs dataset, achieving 98.3% accuracy. Yet, small dataset size and reliance on labelled data limit broader applicability. The study [18] presented multi-modeling and data fusion techniques under IARPA's SCITE program [58], excelling at threat detection but facing scalability and adaptability concerns due to controlled datasets. Sun et al. [116] used COPOD for high-dimensional outlier detection, outperforming Isolation Forest on large heterogeneous datasets. However, comparisons with modern DL approaches and practical integration remain unexplored.

Huang et al. [55] developed ITDBERT, which uses deep temporal-semantic behaviour representation through Word2vec and MaskedLM to capture nuanced temporal sequences. ITDBERT outperforms Deeplog and Isolation Forest in precision and recall but depends on labelled data and high computational resources; future work could explore adversarial sample generation and transfer learning. The paper [75] proposes a method for insider threat detection by analysing both user-specific and role-specific behaviours, addressing limitations in traditional methods that often miss role context, leading to high false positives. Using the unsupervised Isolation Forest algorithm, the study highlights how deviations in user behaviour within similar roles can signal potential threats. Validated over a 17-month dataset, this approach shows improved accuracy over other methods. However, further research is needed to integrate real-time detection and address scalability in larger organisations.

Recent work in insider threat detection has focused heavily on DL and other AI-driven approaches. Models such as the hybrid TCN–Transformer in [127], the stacked CNN–Attentional BiGRU optimised with M-EO in [9], and the spatio-temporal GNN framework SENTINEL in [125] demonstrate strong capability in learning temporal, semantic, and behavioural patterns from large-scale logs. More advanced generative or self-supervised strategies, including diffusion-based architectures like DDCC [28] and LSTM-Diff [117], further enhance detection by addressing data imbalance and sparse anomalies. Transfer learning [68], NLP-based embeddings with LSTMs/GRUs [11], and contrastive representation learning [23] also show how semantic and temporal contexts can be leveraged to achieve superior accuracy and recall across standard benchmarks. Despite impressive performance metrics such as AUC scores exceeding 0.97 in [28] or accuracies above 98% in [56] these techniques face recurring challenges. Many rely on synthetic datasets such as CERT r4.2, r5.2, and r6.2 [9, 11, 68, 117, 127], which, while standardised, lack the realism of enterprise-scale data. High computational costs are another barrier, particularly for complex architectures like adversarial autoencoders in AUTH [133] or graph-based frameworks such as GMFITD [71] and GraphCH [101]. Moreover, robustness drops in data-scarce conditions, as noted in [125], where performance fell by 20% with limited training data. Interpretability also remains limited, reducing

Table 3. DL-Based Detection Approaches

Paper	Overview	Methods/ Models	Features	Limitations
[49]	Uses DL methods with attention mechanisms for insider threat detection via behaviour sequences.	LSTM and Bi-LSTM models with attention mechanism to capture fine-grained user behaviour patterns.	Enhanced performance (3.1%-6.3% improvement), fine-grained detection by analysing historical and sequential behaviour data.	Heavy reliance on CMU-CERT dataset; lacks comparative analysis with broader ML algorithms and real-world scalability testing.
[54]	Introduced ResHybnet, a hybrid model using DL for insider threat detection.	LSTM auto-encoder for feature extraction, GNN and CNN-based elements in a residual hybrid network.	Improved F1-Score by 0.56%, superior detection using GNN for topological connections and CNN for non-topological patterns.	Limited to the CERT 4.2 dataset and lacks solutions for data imbalance problems.
[76]	Introduced a method for insider threat detection using system logs.	Analyses system logs with a recurrent neural network, treating logs as natural language sequences.	Effective for detecting insider threats by identifying patterns in user log sequences, achieving 93% accuracy.	Could benefit from continuous learning strategies to handle diverse behaviours.
[80]	Introduces insider threat detection using multi-autoencoders for handling imbalanced datasets.	Leverages GRUs and autoencoders with multi-head attention for temporal user behaviour analysis.	Outperforms Isolation Forest and LSTM-autoencoders; effective in detecting malicious activities with high accuracy.	Limited exploration of scalability and real-world performance in more diverse data environments.
[89]	DL-based detection of insider threats in enterprise environments.	Stacked-LSTM/GRU models with attention to capture temporal and contextual aspects of user behaviour logs.	Enhanced detection performance compared to basic models; attention mechanism improves model interpretability.	Needs further exploration of scalability and real-time deployment in large-scale environments.
[95]	Focused on creating persona-specific profiles using DL for threat detection.	Utilised FFNN, CNN, LSTM, and ensemble models for profiling user-generated vs. device-generated data.	High F1-Score (98.89%) in detecting cyber-persona types and insider threats; profiles users based on traffic data.	Limited focus on improving DL model performance and context detection for web traffic.
[96]	Proposed DL-based method of identifying insider threats via spectral and spatial analysis of activity logs.	Combines SMOTE-ENN for class imbalance, image-based feature representation using WCNN.	Achieved 97.19% accuracy and high recall through image-based analysis of user activity patterns.	More research needed to understand the role of image-based analytics in cybersecurity.
[129]	Introduces the DAITD framework using GANs to balance datasets and detect insider threats.	Uses LSTM-autoencoder for user behaviour, GANs for synthetic sample generation, and DT/ XGB for classification.	Effectively addresses class imbalance; achieves high F1-Scores (0.9837 for XGB+GAN) and detects multiple insider threat scenarios (CERT dataset).	Requires broader experimentation across more diverse insider threat scenarios and datasets for validation.
[9, 127]	Sequence modelling with CNN/TCN + RNNs.	Hybrid TCN-Transformer; CNN-Attentional BiGRU (M-EO optimised).	Strong capture of temporal + contextual patterns; improved accuracy/recall.	Evaluated mainly on CERT datasets; scalability to real-world enterprise data not tested.
[33, 71, 101, 125]	Graph-based insider detection.	SENTINEL spatio-temporal GNN; GMFITD (GNN + meta-learning); GraphCH (graph with psychological + behavioural features).	Learns relational/temporal structures; supports multimodal signals; handles data scarcity.	Performance drops in data-scarce settings; complex and computationally expensive; limited interpretability.
[28, 117]	Diffusion-based anomaly detection.	DDCC and LSTM-Diff architectures.	Addresses imbalance and sparse anomalies; AUC > 0.97 reported.	High computational cost; reliance on synthetic CERT datasets.
[56, 133]	Autoencoder-based deep learning frameworks.	AUTH adversarial autoencoder; IDU ensemble DL model.	Improves data representation and anomaly detection; >98% accuracy.	Computationally demanding; limited real-world validation.

analyst trust in real-world deployment. Future directions emphasise adaptability, transparency, and multimodal integration. Meta-learning approaches like GMFITD [71] address data scarcity, while user specific decoders in [68] improve personalisation of anomaly detection. GraphCH [101] highlights the potential of combining psychological and behavioural signals with cyber logs, reducing false positives through enriched feature diversity. Contrastive learning [23] and adversarial approaches [133] suggest pathways for unsupervised adaptation across evolving datasets. Coupled with XAI to improve transparency, these AI-based methods form a powerful foundation for scalable, adaptive insider threat detection, though practical deployment will require balancing accuracy, efficiency, and interpretability.

The overview, methods, models, features, and limitations of these DL-based, ML-based, and other AI-based detection approaches are summarised in Tables 3, 4, and 5, respectively.

6.1.2 Statistical Based Detection Approaches. Statistical-based detection approaches identify insider threats by analysing deviations from baseline user behaviour, detecting anomalies that differ from expected actions. However, these methods may struggle to capture complex insider threats without additional context. The study in [36] explores how gender stereotypes impact insider threat assessments, showing that gender biases can distort managerial perceptions of malicious intent. However, its hypothetical nature limits real-world applicability, and it does not fully address underlying mechanisms or solutions. Future research should explore intersecting biases like race and age, emphasising the need to address gender bias in security evaluations.

Table 4. ML-Based Detection Approaches

Paper	Overview	Methods/ Models	Features	Limitations
[27]	Evaluates ML algorithms, focusing on insider threat detection.	Uses KNN, achieving 93.5% accuracy; explores various ML algorithms on the CERT dataset.	Highlights KNN performance for insider threat detection with promising results.	Reliance on a single dataset; lacks comparative analysis with other ML models like RF; limited exploration of real-world deployment challenges.
[60]	Investigates insider threat detection using supervised ML techniques on email logs.	Uses simpler models to avoid overfitting, with Adaboost identified as the most effective algorithm.	Achieves 98.3% accuracy; evaluates several algorithms for threat detection in a limited dataset.	Small dataset limits generalisability; supervised learning depends on labelled data, which may not always be available in real-world cases.
[64]	Proposes a user behaviour analysis system to address data imbalances and detect insider threats.	Uses data adjustment strategies (suspicious data selection, misclassification retention) and ERPSO-XGBoost for modelling.	Superior performance in analysing behavioural characteristics of users, dynamic management strategy adjustments, and insights for enterprise management.	Limited by the use of a single dataset (CERT 4.2), necessitating further exploration of diverse datasets.
[75]	Identifies insider threats by comparing user-specific and role-specific behaviours.	Uses unsupervised Isolation Forest algorithm to detect behaviour deviations relative to peers in the same job role.	Demonstrates robustness in detecting insider threats over 17-month dataset, reducing false positives by focusing on role-specific behaviours.	Further exploration needed into real-time detection and scalability in larger organisations.
[80, 96, 129]	ML-based insider detection with hybrid or ensemble methods.	GAN + XGBoost/Decision Tree; Isolation Forest vs autoencoder-GRU; WCNN with SMOTE-ENN.	Handles imbalance (SMOTE-ENN, GAN); competitive accuracy (97%); interpretable baseline comparisons.	Mostly benchmarked on CERT datasets; limited scalability to enterprise environments; handcrafted features often required.

Table 5. Other AI-Based Detection Approaches

Paper	Overview	Methods/ Models	Features	Limitations
[5]	Focuses on reducing false positives/ negatives in insider threat detection.	Uses feature extraction, sentiment analysis, and Gray encoding to improve detection accuracy with data processing techniques.	Achieves enhanced accuracy (97% AUC) with reduced false positives (0.29%) and false negatives (2.47%) through combined techniques.	Relies on a specific dataset (CERT r4.2) and faces challenges in real-time threat detection scalability due to Gray encoding complexity.
[18]	Detect insider threats, sponsored by IARPA SCITE.	Combines multiple models and data fusion techniques; highlights performance on complex real-world problems.	Strong performance across various metrics; explores multi-modeling and data fusion to improve insider threat detection.	Relies on controlled datasets, raising concerns about scalability and adaptability in real-world contexts.
[55]	Introduces ITDBERT using Word2Vec and MaskedLM for detecting insider threats.	Learns user behaviour representation for insider threat using Word2vec and MaskedLM. Combines ITDBERT model with dual granularity temporal information and Bi-LSTM for behaviour detection.	High F1-Score and AUC in behaviour detection; improved precision and recall compared to Deeplog and Isolation Forest.	Dependency on labelled data and high computational resources; pre-training and implementation limitations in smaller organisations.
[116]	Insider threat detection by handling large-scale data.	Copula-Based Outlier Detection for fast, multi-dimensional data processing without parameter tuning.	Outperforms Isolation Forest in processing heterogeneous data quickly and efficiently.	Lacks comparison with novel anomaly detection techniques, exploration on scalability and security framework integration.
[132]	Introduced a heterogeneous graph embedding model for insider threat detection.	Uses Word2Vec and heterogeneous graphs for mapping user behaviour into high-dimensional space.	Achieved an AUC of 0.98 and F1-Score of 0.89; handles user relationships and sequence information effectively.	Needs further investigation for generalisation beyond a specific dataset.
[68]	Transfer learning for personalization.	Pretrained embeddings with user-specific decoders.	Adapts to new users and contexts; improves personalization.	Requires large pretrained corpora; high computational demand.
[11, 23]	NLP and representation learning strategies.	NLP-based embeddings with LSTM/GRU; contrastive learning for behaviour modelling.	Captures semantic and contextual nuances; boosts accuracy and recall.	Interpretability challenges; dataset dependency.

The paper [103] proposes a model combining deterrence theory, **Situational Crime Prevention (SCP)**, and planned behaviour theory to address insider threats. It highlights how perceived sanctions, increased effort, and reduced rewards influence security attitudes, with employee survey data used for model validation. However, reducing provocations had limited effect, possibly due to cultural factors. The model emphasises that secure behavioural intentions are shaped by attitudes, control, and norms, suggesting the need for refinement in diverse contexts. A limitation of the study is the subjectivity associated with self-reported data used for validation. Another study [104] integrates Crime Prevention and Social Bond Theory, showing that increasing effort and risk, reducing rewards, and removing excuses deter misconduct. Statistical analysis highlights that commitment to policies and involvement in security activities lower intentions for malicious actions. While offering practical strategies, the study underscores the need for further research on psychological and cultural influences in security, providing a novel perspective on insider threat mitigation.

The study [126] proposes an insider threat detection method combining statistical and sequential analysis through the **Convolutional Attention and Transformer Encoder (CATE)** framework. This dual approach integrates CNN-based feature extraction with transformers for temporal dependencies, improving detection rates and reducing false positives on the CERT dataset. However, it

Table 6. Statistical Based Detection Approaches

Paper	Overview	Methods/ Models	Features	Limitations
[8]	Investigates insider threat detection using log analysis and probabilistic event correlation.	Utilises log management and Bayesian techniques for probabilistic event correlation in detecting insider threats.	Emphasises real-time log analysis and proposes a probabilistic approach to balance detection accuracy with low false alarms.	Scalability issues and challenges in adapting the Bayesian model across diverse network architectures. Lacks a detailed comparison of tools.
[29]	Propose a model to predict insider threats by modelling risks using historical data.	Bayesian Network model integrating technological, organisational, and human factors.	Robust methodology incorporating various risk factors, but primarily focused on technological aspects of insider threat prediction.	Heavy reliance on historical data, which can introduce biases, and a focus on technology over human behaviour and organisational culture.
[36]	Explores the impact of gender bias on insider threat assessments, using controlled scenarios.	Examines gender stereotypes and chivalry hypothesis through controlled scenarios in information system security.	Highlights the influence of gender bias on managerial assessments of insider threats.	Limited applicability of findings due to hypothetical scenarios; does not deeply explore interventions or underlying mechanisms of biases.
[103]	Introduces a deterrence and prevention-based model to mitigate insider threats in organisations.	Combines deterrence, crime prevention, and behavioural theories with surveys analysed using statistical methods.	Comprehensive model incorporating sanction certainty, effort/risk increase, and reward reduction to influence security attitudes.	Cultural factors may reduce the effectiveness of reducing provocations and excuses. Further exploration of additional factors is needed.
[104]	Proposes a motivation and opportunity based model to reduce information security insider threats in organisations.	Combines crime prevention and social bond theories with statistical analysis of employee security behaviour.	Demonstrates that commitment to policies and involvement in security activities reduce insider threat risks.	Cultural factors may weaken the impact of reducing provocations on employee behaviour, requiring further investigation.
[126]	Proposes a framework for insider threat detection, combining statistical and sequential analysis.	Uses CATE framework for feature extraction and capturing temporal dependencies.	Combines CNNs for statistical feature extraction and Transformer models for temporal data, improving detection rates and reducing false positives.	Dependency on high-quality sequence data, and challenges in adapting to dynamic threats.

relies on high-quality user sequence data and faces challenges with dynamic threats. Future work could focus on finer data granularity and multi-granularity fusion for low-frequency threats. The study [8] examines insider threat detection through log analysis and event correlation, proposing a probabilistic event correlation approach for balancing accuracy and false alarm rates. Although it discusses existing tools, it lacks comparative analysis across settings, and the Bayesian approach may face scalability issues across varied networks. Future research could add preventive measures for better practical applicability. The authors in [29] present a BN method that integrates technological, organisational, and human factors for predicting insider threats. However, its reliance on historical data limits its ability to detect new threats and is dependent on high-quality data. Future work should incorporate real-time data and adaptive learning for a more comprehensive prediction system.

The overview, methods and models, features, and limitations of these statistical-based detection approaches are summarised in Table 6. Some of the studies discussed above have employed purely statistical methods to demonstrate relationships between different factor variables, but human factors that contribute to insider threats can still be observed. Although some studies have integrated statistical approaches with technical approaches to improve accuracy and robustness, they have remained limited to technical measures to study on behaviours while overlooking human aspects. Therefore, there is still a need to incorporate human factors into technical solutions.

6.1.3 Manual and Heuristic-Based Detection Approaches. These approaches rely on human expertise and predefined rules to identify suspicious activities. Manual detection involves security analysts monitoring user behaviours and system logs based on knowledge of insider threat patterns. Heuristic-based methods use expert-crafted rule-based algorithms to define abnormal behaviour. While effective in predictable environments, these approaches may lack scalability and adaptability for detecting sophisticated or evolving attacks. The study [61] examines how opportunity-reducing strategies based on SCP theory can mitigate insider threats. It explores techniques to increase perceived effort and risks while reducing rewards and excuses. Furthermore, the study emphasises the need to balance “hard” measures, such as strict surveillance that may heighten discomfort and risk perception, with softer preventive approaches. It concludes by calling for further research into the psychological effects of security measures and the identification of an optimal balance between control and acceptability.

The study in [13] extends the **Role-Based Access Control (RBAC)** model by integrating risk assessment and trust evaluation, introducing a method to compute risk values for permissions and

Table 7. Manual and Heuristic-Based Detection Approaches

Paper	Overview	Methods/ Models	Features	Limitations
[12]	Proposes SAAF to dynamically adjust access control in response to threats.	Dynamic adaptation of access control policies, feedback loop for continuous updates based on violations.	Effective mitigation of insider threats, adaptive system responding to malicious behaviours in real-time.	Scalability issues, limited by case studies and predefined scenarios not representing the complexity of real-world environments.
[13]	Expands RBAC model with risk assessment and trust to mitigate insider threats.	Role activation algorithm, risk assessment, and optimisation problem for minimizing risk exposure.	Simulation strategy to detect inference threats and heuristic evaluation for minimizing risk during role selection.	Limited exploration of integration for real-time monitoring and enhancement needed.
[37]	Insider detection via deception technology.	Honeytokens, honey accounts, decoys deployed in enterprise (healthcare) systems.	Provides high-fidelity alerts, low noise, rapid incident response; human-centric traps attackers cannot easily bypass.	Requires careful design/placement; scalability challenges; limited coverage of routine insider behaviours; potential operational noise if misused.
[61]	Investigates opportunity-reducing strategies in insider threat prevention.	SCP-based categorisation of security techniques; scenario-based surveys and PLS-SEM data analysis.	Highlights the impact of hard and soft techniques on employee behaviour, balancing effort, risk, rewards, and excuses in security policies.	Limited exploration of the long-term psychological impact of security measures, lack of focus on balancing soft and hard techniques for long-term security and morale.

roles. It formulates an optimisation problem aimed at minimising risk exposure and uses a role activation algorithm to provide a solution. Simulations help detect inference threats before policy deployment. The system can deny access to high-risk permissions, but further work is needed to integrate monitoring. The paper [12] presents the **Self-Adaptive Authorisation Framework (SAAF)**, which adjusts access control policies to counter insider threats. While promising, SAAF's reliance on case studies and simulations limits its real-world applicability, and scalability challenges may arise in live environments. Future research should test SAAF in real-world settings to better assess its effectiveness.

The experience study in [37] highlights how honeytokens, honey accounts, and decoys can effectively expose malicious insiders in real-world deployments. Unlike algorithmic models, deception leverages human-centric traps that attackers cannot easily avoid once engaged. Its demonstrated success in a healthcare setting shows that well-designed deception artifacts can deliver high-fidelity alerts, low noise, and rapid incident response. Nevertheless, deception has inherent limitations. The effectiveness of honeytokens or decoys depends on careful placement and design; poorly executed artifacts risk detection or may even create operational noise if legitimate users interact with them. As [37] notes, scalability is another issue, since maintaining convincing artifacts across diverse enterprise systems requires significant manual effort. Compared to AI-based models, deception does not cover the full spectrum of insider behaviours, particularly when malicious actions mimic routine patterns without touching decoys. To maximise its utility, deception should evolve in tandem with ML. Integration with AI-driven methods could enable automated lure generation, adaptive attack-path mapping, and real-time validation against evolving attacker techniques. As suggested in [37], coupling deception with UEBA or anomaly detection models could provide a layered defence that balances interpretability and accuracy. In this way, deception serves as a valuable heuristic tool, most effective when combined with statistical or AI-based detection to form a robust defence-in-depth strategy.

The overview, methods, models, features and limitations of these manual and heuristic-based detection approaches are summarised in Table 7.

6.1.4 Hybrid Detection Approaches. Hybrid detection approaches combine rule-based methods, statistical analysis, and ML to improve insider threat detection accuracy. By integrating different techniques, they address the limitations of individual methods. The study in [97] presents a framework that combines advanced analytics with policy regulation, using a bio-inspired design for proactive threat management. While formal verification of access control policies enhances rigour, practical implementation and scalability need further exploration. The use of unsupervised ML for anomaly detection is innovative, but distinguishing benign anomalies from true threats may be

challenging in complex environments. Therefore, more extensive testing is needed to validate the effectiveness of the framework.

A hybrid framework called Employee-Watcher was developed in [98], where insider threats are detected using a combination of ML and statistical methods. It includes three modules such as data collection, feature engineering, and anomaly checking. The data collection module aggregates user activity, which is then preprocessed. Feature engineering enhances data for analysis, while the anomaly checker uses entropy-based metrics and **Local Outlier Factor (LOF)** to detect unusual behaviour. Experiments show accuracy between 96.1% and 99.7%. Future work could involve testing on diverse datasets beyond CERT r4.2. Another study [30] examines the CITD tool's deployment, emphasising the need for adaptable security policies. While insightful, its reliance on qualitative data and challenges with large datasets may limit scalability and generalisability.

The study [46] proposes using **Gaussian Mixture Models (GMM)** and sensitivity profiles to detect insider threats, combining automated anomaly detection with human expertise to reduce false positives. Visual analytics help security analysts understand anomalies, blending technical and non-technical indicators. However, synthetic data evaluation limits real-world applicability, and sensitivity profiles need further validation. Future work should apply the model to real datasets, enhance scalability, and improve automation. Behavioural and personality traits were analysed within a game-based simulation to predict insider threats in the study [15]. While using real-time data, facial expressions, and linguistic features, its small sample size and simulated setting limit generalisability. The study [121] uses **Alternate Reality Games (ARGs)** for evaluating insider threat detection systems. However, the artificial nature of ARGs and the limited sample size may not fully capture real-world complexities, and the sandbox network might miss subtle security nuances.

The framework produced by Lavanya et al. [69] exemplifies this by integrating Improved PCA, Enhanced BiGAN, and a deep neural network with Bayesian optimisation, thereby addressing data imbalance while optimising hyperparameters. Results show strong accuracy (98.8%) and low false positive rates, outperforming DL-only baselines. Meanwhile, MetaPoL [44] contributes not as a detection system but as a **Virtual Reality (VR)**-based data generation platform, creating realistic, customisable patterns-of-life datasets for nuclear security insider threat modelling. Together, these works highlight the diversity of hybrid strategies, spanning both algorithmic integration and data enrichment. Yet, hybrid approaches introduce new challenges. The computational complexity of multi-stage models such as [69] can hinder deployment in real-time or resource-constrained environments. For MetaPoL [44], limitations stem from the fidelity of VR simulations and reliance on adoption of immersive platforms, raising questions about scalability and realism. Both approaches, though powerful, highlight the inherent tradeoff between robustness and operational feasibility, particularly when extending beyond controlled experimental settings.

Future directions for hybrid detection involve tighter integration of complementary methods. Data-driven AI can be paired with heuristic or statistical preprocessing for better interpretability and efficiency, as seen in [69]. VR-based platforms like [44] could evolve through multi-user interaction, dynamic **Non-Player Character (NPC)** behaviours, and links to real facility sensor data, making simulations even more realistic. By bridging statistical rigor, AI adaptability, and heuristic insights, hybrid approaches promise to deliver balanced, scalable insider threat detection systems with enhanced resilience across diverse environments.

The overview, methods and models of the approach, features and limitations of these hybrid detection approaches are summarised in Table 8.

6.2 Comparison of Detection Techniques

When comparing different techniques used to detect insider threats in the literature, a significant number of papers have employed AI, ML, and DL-based approaches 64%. The use of statistical-based

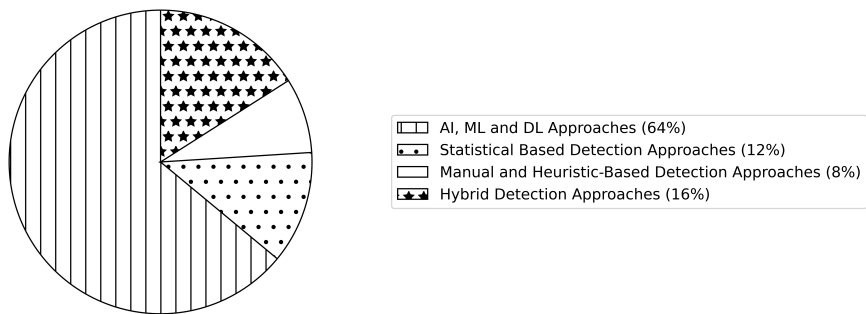


Fig. 3. Insider threat detection techniques comparison.

Table 8. Hybrid Detection Approaches

Paper	Overview	Methods/ Models	Features	Limitations
[15]	Predicts insider threats by analysing behaviour and personality traits in a game-based simulation.	Uses real-time behavioural data, facial expressions, linguistic features, and ensemble learning algorithms.	Potential in balancing precision and recall with real-time data; integrates multiple behavioural traits.	Simulated environment limits real-world applicability; a small sample size reduces generalisability.
[30]	Examines the efficacy of the CITD tool in real-world settings, focusing on insider threat detection challenges.	Qualitative data through industry trials and customised policy implementations.	Insights into practical insider threat detection dynamics, adaptable to varying organisational environments.	Relies on qualitative data from security analysts and limited scalability/generalisability across organisations and industries.
[44]	VR-based data generation platform (MetaPol) for insider modelling.	Virtual reality simulation of customisable "patterns-of-life" datasets for nuclear security scenarios.	Produces realistic, flexible synthetic datasets; supports scenario-driven evaluation of insider detection.	Limited by fidelity of VR simulations; requires immersive platform adoption; questions on scalability and realism in operational settings.
[46]	Uses Gaussian Mixture Models and sensitivity profiles for anomaly detection and reduced false positives.	Combines automated detection with human expertise and visual analytics.	Enhanced detection accuracy by integrating heuristic knowledge and technical/non-technical indicators.	Evaluation based on synthetic data; real-world complexities are not fully represented. Further validation with actual data is needed.
[69]	Hybrid insider threat detection framework.	Improved PCA + Enhanced Bi-GAN + Deep Neural Network with Bayesian optimisation.	Balances data imbalance, optimises hyperparameters; achieves 98.8% accuracy with low false positives; outperforms DL-only baselines.	Computationally complex; resource-intensive, limiting real-time deployment.
[97]	Combines threat analytics with policy regulation using formal methods for verifying access control policies.	Bio-inspired design, real-time state transitions, unsupervised machine learning for anomaly detection.	High rigour in access control policy verification, proactive stance on insider threat management.	Practical implementation and scalability across varied settings need more investigation; difficulty distinguishing benign anomalies from true threats in dynamic environments.
[98]	A hybrid insider threat detection framework combining machine learning and statistical methods.	Uses modules for data collection, feature engineering, and anomaly checking with One-Hot encoding and LoF method.	High detection accuracy (96.1%-99.7%), integrates multiple data sources, and hybrid anomaly detection using entropy metrics.	Limited dataset (CERT r4.2) and scope for improvement in dataset diversity.
[121]	Uses ARGs to evaluate insider threat detection systems in simulated environments.	Employs sandbox networks and controlled, gamified settings to simulate insider threats.	Unique approach simulating insider threats, using ARGs to mirror real-world behaviours.	Artificiality in simulations may not fully capture real insider threat complexities; limited sample size and generalisability.

approaches has 12% proportion and hybrid approaches has a proportion of 16% , but these percentages are lower compared to AI, ML, and DL-based approaches. A minimal number of studies 8% have utilised manual and heuristic-based detection approaches, indicating a declining trend in using such techniques for insider threat detection. However, most studies have focused on intentional insider threats rather than unintentional insider threat detection. The analysis of the detection techniques is illustrated in Figure 3. Overall, the results show a clear trend towards AI, ML, and DL-based approaches, likely due to recent technological advancements. However, these approaches are entirely based on data that contain technical measures, without incorporating human aspects. Therefore, approaches that integrate human factors alongside technical data are essential for more effective insider threat detection. Future research should focus on combining human factors with technical data and generating datasets that can be used to detect insider threats using state-of-the-art AI, ML, or DL methods. Alternatively, new approaches could be developed to

detect insider threats based on both human and technical aspects. In particular, unintentional insider threat detection requires further research and new solutions to effectively address this challenge. Such methods can guide future researchers in creating more accurate and reliable solutions.

6.3 Insider Threat Detection Challenges

The detection of insider threats presents several significant challenges. Although training ML or DL models with balanced datasets typically produces better performance, in practice, malicious insider activities are exceedingly rare compared to benign behaviours. Moreover, collecting substantial volume of malicious insider activities is also impractical, leading to highly imbalanced insider threat datasets [73, 74]. Additionally, there is a lack of proper testbeds since there are no publicly available real-world datasets except the CERT dataset [42], which still has discrepancies between synthetic data and real-world scenarios [130].

Simply detecting attacks based on activity types is insufficient because the same activity can be either benign or malicious, highlighting the importance of temporal information in analysing user behaviour to identify threats. However, most current approaches focus solely on activity types and struggle to incorporate temporal information. Leveraging and fusing various data sources is crucial for improving insider threat detection, but existing work typically treats it as an anomaly detection task, viewing anomalous samples as out-of-distribution. In reality, insiders' behaviour may not significantly deviate from the norm, making them hard to distinguish from benign users. Insiders often refine their strategies to evade detection, necessitating adaptable models that can enhance detection performance [111, 131].

Current methods primarily detect insider threats after malicious activities have occurred, causing significant damage. Thus, early detection of insider threats is a critical topic. Despite the increase in insider threat incidents, many organisations do not report these incidents or share data due to ethical and privacy concerns, limiting access to valuable data. Offline detection tools also fall short as they cannot provide real-time support, hindering timely responses. Insiders with trusted access can bypass external network security tools like firewalls, intrusion detection systems, and antivirus software, presenting a significant challenge for existing tools. Moreover, traditional AI-based models often lack interpretability, making it essential to provide insights to domain experts to take confident actions [2, 35]. Therefore, these challenges must be addressed when developing an effective insider threat detection mechanism.

7 Datasets Used in Insider Threat Detection

In recent years, insider threat research has used existing datasets, given that the development of new datasets is both resource intensive and time consuming. Such datasets, which commonly integrate normal activity with synthesised anomalies, are integral to threat categorisation and the development of automated detection models. This section examines the principal datasets employed within the literature.

The Lincoln Laboratory dataset [17] was developed by MIT Lincoln Laboratory, to evaluate the **Defence Advanced Research Projects Agency (DARPA)** intrusion detection system and it was created and recorded on a simulated United States Air Force facility network, connected to the Internet. The dataset includes 32 types of attacks and those attacks are grouped into four types: (1) denial of service, (2) user to root, (3) remote to user, and (4) probes. When creating the dataset, system logs were collected and each log contains information such as system calls, execution date and time, process details, arguments, user ID, and group ID. Researchers have used this dataset to study methods for detecting malicious insiders.

Realistic logs of masquerader behaviour were collected through an experiment [105], and these were subsequently used to construct the RUU dataset [66]. A total of 34 normal users, along with

another 14 users who mimicked as masqueraders to find data with financial value, were involved when generating the dataset.

The Enron dataset [67] consists of 500,000 e-mails from 150 Enron employees, mainly senior management and it is a traitor based dataset. Due to privacy requests, attachments were removed, and some e-mails were deleted. While originally a resource for e-mail research, it is now also used for insider threat detection, particularly for text and social network analysis.

The Schonlau dataset [109] contains 15,000 Unix shell commands for 50 users, with 20 users simulating masquerade activities. It is widely used in academic research on masquerader detection. The first 5,000 commands for each user are clean and serve as training data, while the remaining commands include masquerade data with a 5% probability. However, the dataset only provides Unix command sequences and user names, without additional details such as flags, timestamps, or arguments, which presents some limitations.

Timing and positioning data of mouse movements from 10 users were collected and stored in training files in the Balabit mouse dynamics challenge dataset [32]. The purpose of the challenge is to protect users from unauthorised account access. The dataset includes fields such as record timestamp, client timestamp, button state, and x and y coordinates. This dataset has used to study mouse bio behavioural features for insider threat detection.

The Greenberg's dataset [67] is an authentication-based dataset containing full Unix C shell commands from 168 users. Unlike the Schonlau dataset, it includes arguments and timestamps in the command entries. For insider threat detection, some users are randomly selected to act as masqueraders.

The TWOS dataset [47] was collected from real user interactions with a host machine, containing both legitimate user data and instances of malicious insiders including masqueraders and traitors. It was gathered during a competition held by the Singapore University of Technology and Design in March 2017. The dataset includes six types of data sources: keystrokes, mouse activity, host monitor, network traffic, SMTP logs, and logon data, along with additional information from a psychological personality questionnaire.

Among the publicly available insider threat resources, the CERT dataset is the most popular and widely used. The CERT dataset [22] is a synthetic dataset containing system logs with labelled insider threat activities. It is widely used in recent DL-based insider threat detection studies. This is maintained by the CERT division at **Carnegie Mellon University (CMU)**, and it is based on over 1,000 real insider threat cases and includes synthetic datasets with scenarios of traitors and masqueraders. The dataset contains five log files tracking computer-based activities of employees in a simulated organisation. (1) logn.csv: records logon and logoff events, (2) email.csv: logs all email activities (send, receive), (3) http.csv: tracks web browsing (visit, download, upload), (4) file.csv: logs actions with removable media (open, write, copy, delete), (5) device.csv: records thumb drive usage (connect, disconnect). Additionally, the dataset includes a psychometric file, psychometric.csv, containing the "Big Five" personality traits for each employee. The data is generated using various models, such as social, content, and psychometric models, to simulate realistic behaviours and workplace dynamics.

A summary of above discussed datasets is illustrated in Table 9.

8 Discussion

The analysis addressed four guiding RQs and provides insights that clarify the state of the field and highlights its key limitations and research opportunities.

First, the literature consistently distinguishes intentional from unintentional insider threats on the basis of motivation and intent. Malicious insiders act deliberately to harm organisations, whereas unintentional insiders create risk through negligence, lack of awareness, or simple human

Table 9. Summary of Insider Threat Datasets

Dataset	Type of Threat	Data Collected	Use Cases	Limitations
Lincoln Laboratory Intrusion Detection Dataset [17]	Masqueraders and Traitors	System logs (e.g., system calls, process details, user ID, group ID), 32 attack types grouped into 4 categories: denial of service, user to root, remote to user, probes	Evaluation of intrusion detection systems, malicious insider detection	Simulated network data, might not reflect real-world variations
Are You You (RUU) Dataset [105]	Masqueraders	Logs of masquerader behaviour from 34 normal users and 14 masqueraders	Study of masquerader detection	Limited number of users, focused on financial-value data
Enron Dataset [67]	Traitors	500,000 emails from 150 employees (primarily senior management); attachments and some emails removed due to privacy requests	Email research, insider threat detection, text and social network analysis	Lack of attachments, email deletions, privacy concerns
Schonlau Dataset [109]	Masqueraders	15,000 Unix shell commands from 50 users; 20 users simulate masquerade activities	Research on masquerader detection	Lacks additional details such as flags, timestamps, or arguments, limiting depth of analysis
Balabit Mouse Dynamics Dataset [32]	Account Hijacking	Timing and positioning data of mouse movements (x and y coordinates, button state, timestamps) from 10 users	Study of mouse bio-behavioural features for insider threat detection	Focused on mouse movements only, no other user activity captured
Greenberg's Dataset [67]	Masqueraders	Unix C shell commands from 168 users, includes arguments and timestamps	Insider threat detection based on command-line activity	Limited to Unix command-line data
TWOS Dataset [47]	Masqueraders and Traitors	Six data types: keystrokes, mouse activity, host monitor, network traffic, SMTP logs, logon data, and psychological questionnaire	Multi-source insider threat detection, competition-based data	May not fully capture real-world dynamics
CERT Dataset [22]	Masqueraders and Traitors	System logs (logon, email, web activity, file actions, device usage) and psychometric data ("Big Five" personality traits)	Deep learning studies, realistic insider threat scenarios (based on over 1,000 real cases)	Synthetic dataset, might not fully represent real-life complexities

error. While this distinction appears in many taxonomies and models, the review reveals that insights relevant to UITs are comparatively scarce. The unpredictability of human error, coupled with the difficulty of attributing intent, contributes to a relative neglect of UITs in technical detection research. This underlines the need for focused study of UITs as a distinct class of security problem.

Second, the review synthesised how existing research characterises human factors contributing to insider threats. Using the review a novel taxonomy was developed categorising these factors into individual (e.g., cognitive load, stress, personality traits), situational/task-related (e.g., time pressure, task complexity, poor procedures) and organisational (e.g., culture, policies, leadership) dimensions. Such structuring provides a clearer framework for future modelling where behavioural and contextual elements are more predictive than intent.

Third, the review identified both strengths and limitations of current detection techniques. Advances in ML and DL have yielded strong performance in benchmarked tasks, particularly for recognising malicious behaviours in technical data. Most studies focus on intentional insider threats, leaving unintentional threats under explored by heavily relying on technical data, overlooking human factors. Future research should integrate behavioural and contextual elements with technical measures to enable more accurate and reliable detection. Yet, the existing insider threat detection models report strong performance metrics, particularly in terms of F1-scores and AUC values. DL based approaches consistently achieve high accuracy and robustness, with models such as ResHybnet attaining an F1-score of 92.19%, while advanced frameworks like DAITD further push performance to an average F1-score above 0.98. Similarly, approaches using WCNN and heterogeneous graph embeddings report AUC values around 0.97–0.98, highlighting their capability to capture complex patterns. Persona based profiling and ensemble models also demonstrate strong performance, reaching F1-scores and AUC values exceeding 0.95 in real world data.

Many studies report high performance in controlled settings, with F1 and AUC values frequently exceeding 0.90 on benchmark datasets such as CERT [5, 27, 54, 76, 96, 129]. However, closer examination shows that these improvements are often marginal, with gains over established baselines such as SVM, RF, and CNN typically amounting to fractions of a percent. Substantial

advances are usually linked to data augmentation or representation learning techniques (e.g., GANs, autoencoders, graph embeddings), which help mitigate class imbalance and improve recall for rare insider behaviours. While these strategies can raise F1-scores by balancing false positives and false negatives, their reliance on narrow datasets restricts generalisability, and inflated results on CERT may not transfer to heterogeneous, real-world environments. Evaluation practices further complicate comparison: some studies report accuracy, others F1 or AUC, and only a minority include precision and recall. This inconsistency obscures weaknesses, particularly in minority class detection. Moreover, even very high benchmark scores do not guarantee operational readiness, as critical issues including scalability, real time applicability, false positive burden, interpretability, and cross organisational generalisability remain insufficiently addressed. Overall, while detection models show promising benchmark performance, their dependence on limited datasets and technical features highlights the need for cautious interpretation. Future research should validate results across diverse datasets and integrate human and contextual factors to enhance robustness in practical deployments.

However, integrating LLMs to both generate synthetic logs or modelling scenarios of insider threats and to detect such threats is emerging [34]. Systems that continuously score insider risk, adjust behaviour baselines, and update detection policies in near real time. These often involve auto encoders, anomaly detection networks, or experts in the loop to refine what “normal” behaviour is [120]. Even though emerging approaches show promise, several challenges remain. Insider behaviour is dynamic and influenced by organisational policies and work environments, which necessitates more federated and adaptive solutions. More broadly, the generalisability of current models across different contexts is limited, and achieving real time threat detection is computationally costly. Furthermore, human factors such as stress, fatigue, and organisational culture are frequently overlooked. Incorporating these dimensions could enhance detection, particularly for unintentional threats, yet the associated data are difficult to collect, quantify, and integrate.

Finally, an analysis of datasets demonstrates that most ML approaches rely heavily on technical traces such as logs and e-mails. While datasets like CERT and Enron have driven progress, they are synthetic or domain-specific and rarely incorporate meaningful human attributes. The scarcity of integrated human-technical datasets restricts the development of models capable of accounting for human factors, a critical requirement for progress in UIT detection. Although some CERT variants include psychometric features, these remain exceptions rather than the norm, underscoring the urgent need for new data collection and sharing mechanisms.

Overall, the analysis underscores a significant shortcoming in insider threat research: the insufficient integration of human and contextual factors with technical evidence. Although technical models achieve strong results on benchmark datasets, their reliance on synthetic data and disregard for human dimensions limit their applicability, particularly in detecting unintentional threats. Addressing this limitation will require the creation of richer human-technical datasets and the adoption of modelling approaches that incorporate behavioural, organisational, and contextual dynamics alongside technical indicators.

8.1 Threats to Validity of the Review

Although this systematic review adhered to a well-established methodology [88], the results may not comprehensively cover all relevant studies due to certain limitations in the review process. Therefore, this section addresses potential threats to validity and the steps taken to mitigate them, focusing on construct, internal, external, and conclusion validity.

Threats to construct validity stem from the reliance on search term queries and the coverage of selected repositories, which may have led to the omission of relevant studies. This risk was mitigated by employing Google Scholar in a snowballing process, although some publications

may still be missing. Errors in applying inclusion and exclusion criteria also pose a threat, but cross-checking against the primary studies was conducted to reduce this risk.

Internal validity may be affected by potential errors in data extraction and analysis. To mitigate this, cross checking was performed and consensus reached among authors before finalising the results, though minor errors may still persist. External validity is limited by the review's focus on studies from 2019 to October 2025, which may exclude earlier work and overlook temporal variations in trends. Bias may arise from restricting the search to English language papers, reviewer subjectivity, and the general tendency to publish positive results. These risks were mitigated through a peer verified process, cross-checking, and active involvement of all authors in study selection and review.

9 Research Opportunities and Conclusion

This review has analysed the current state of insider threat research with a specific focus on the role of human factors. Taken collectively, the literature demonstrates notable technical development in insider threat detection; however, it also reveals enduring weaknesses in the conceptualisation and measurement of human behaviour. The most salient insight is that progress in the field remains constrained less by the lack of modelling techniques and more by the limited availability of realistic, behaviour-centric data. Despite the proliferation of detection approaches, the dominant reliance on synthetic datasets and constrained experimental settings restricts generalisability and limits the operational relevance of existing solutions.

A key contribution of this review is the identification of a shared gap across both malicious and unintentional insider threat research. While malicious insiders have received extensive analytical attention, unintentional insider threats remain comparatively underexplored despite their prevalence in real-world incidents. The synthesis presented here underscores the need for approaches that integrate behavioural, organisational, and contextual variables into detection pipelines. In particular, incorporating human factor information during dataset construction and model design offers a critical pathway to more comprehensive and accurate threat identification.

At a methodological level, the findings highlight the potential of hybrid detection frameworks, reinforcement learning, ontological modelling, and privacy-preserving approaches such as federated learning. These techniques offer promising solutions to current limitations regarding data scarcity, scalability, and cross-organisational deployment. Nevertheless, the review recognises inherent limitations in the existing body of work, including the heterogeneity of datasets, inconsistencies in reporting standards, and the absence of large-scale empirical validation. These constraints reflect broader challenges in insider threat research and should be acknowledged in the interpretation of current evidence.

The practical implications of this synthesis are clear: technological countermeasures alone are insufficient for mitigating insider threats. Effective solutions require a socio-technical perspective that combines detection mechanisms with training, organisational policy development, and interventions aimed at reducing human-factor-driven vulnerability. As insider threats increasingly emerge from complex interactions between individuals, systems, and organisational contexts, security strategies must evolve accordingly.

Looking forward, the field is well-positioned for advancement. Future research that integrates behavioural science, adaptive AI techniques, Large Language Models and ontology-driven knowledge representation can support the development of more predictive, transparent, and generalisable systems. Achieving this will require sustained collaboration between academia and industry to generate richer real-world datasets and evaluation environments. By addressing these research directions, the community can move toward a more comprehensive and proactive approach to insider threat mitigation, advancing both theoretical understanding and practical resilience in contemporary organisational settings.

References

- [1] Mohammed Nasser Al-Mhiqani, Rabiah Ahmad, Zaheera Zainal Abidin, Warusia Mohamed Yassin, Aslinda Hassan, Ameera Natasha Mohammad, and Nathan L. Clarke. 2018. A new taxonomy of insider threats: An initial step in understanding authorised attack. *International Journal of Information Systems and Management* 1, 4 (2018), 343–359.
- [2] Mohammed Nasser Al-Mhiqani, Rabiah Ahmad, Z. Zainal Abidin, Warusia Yassin, Aslinda Hassan, Karrar Hameed Abdulkareem, Nabeel Salih Ali, and Zahri Yunus. 2020. A review of insider threat detection: Classification, machine learning techniques, datasets, open challenges, and recommendations. *Applied Sciences* 10, 15 (2020), 5208.
- [3] Khuloud Saeed Alketbi and Abid Mehmood. 2025. A comprehensive survey of explainable artificial intelligence techniques for malicious insider threat detection. *IEEE Access* 13 (2025), 121772–121798. DOI: [10.1109/ACCESS.2025.3587114](https://doi.org/10.1109/ACCESS.2025.3587114)
- [4] Maher Alsharif, Shailendra Mishra, and Mohammed AlShehri. 2022. Impact of human vulnerabilities on cybersecurity. *Computer Systems Science & Engineering* 40, 3 (2022), 1153–1166. DOI: [10.32604/csse.2022.019938](https://doi.org/10.32604/csse.2022.019938)
- [5] Muhanned Alslaiman, Mohammed I. Salman, Mariam M. Saleh, and Bin Wang. 2023. Enhancing false negative and positive rates for efficient insider threat detection. *Computers & Security* 126 (2023), 103066. DOI: [10.1016/j.cose.2022.103066](https://doi.org/10.1016/j.cose.2022.103066)
- [6] Fatima Rashed Alzaabi and Abid Mehmood. 2024. A review of recent advances, challenges, and opportunities in malicious insider threat detection using machine learning methods. *IEEE Access* 12 (2024), 30907–30927. DOI: [10.1109/ACCESS.2024.3369906](https://doi.org/10.1109/ACCESS.2024.3369906)
- [7] Fatima Rashed Alzaabi and Abid Mehmood. 2024. A review of recent advances, challenges, and opportunities in malicious insider threat detection using machine learning methods. *IEEE Access* 12 (2024), 30907–30927. DOI: <https://doi.org/10.1109/ACCESS.2024.3369906>
- [8] Amruta Ambre and Narendra Shekhar. 2015. Insider threat detection using log analysis and event correlation. *Procedia Computer Science* 45 (2015), 436–445. DOI: [10.1016/j.procs.2015.03.175](https://doi.org/10.1016/j.procs.2015.03.175)
- [9] A. Anju and M. Krishnamurthy. 2024. M-EOS: Modified-equilibrium optimization-based stacked CNN for insider threat detection. *Wireless Networks* 30, 4 (2024), 2819–2838.
- [10] Anju A, Nithya Kalyani M, Shalini K, Haritha Ravikumar, Saranya P, and M. Krishnamurthy. 2023. Detection of insider threats using deep learning. In *Proceedings of the 2023 3rd International Conference on Pervasive Computing and Social Networking (ICPCSN)*. IEEE, 264–269. DOI: [10.1109/ICPCSN58827.2023.00050](https://doi.org/10.1109/ICPCSN58827.2023.00050)
- [11] Neda Baghalizadeh-Moghadam, Christopher Neal, Frédéric Cuppens, and Nora Boulahia-Cuppens. 2024. NLP and neural networks for insider threat detection. In *Proceedings of the 2024 IEEE 23rd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*. IEEE, 2010–2018.
- [12] Christopher Bailey and Rogério de Lemos. 2020. Malicious changeload for the resilience evaluation of self-adaptive authorisation infrastructures. *Future Generation Computer Systems* 113 (2020), 113–131. DOI: [10.1016/j.future.2020.06.045](https://doi.org/10.1016/j.future.2020.06.045)
- [13] Nathalie Baracaldo and James Joshi. 2013. An adaptive risk management and access control framework to mitigate insider threats. *Computers & Security* 39 (2013), 237–254. DOI: [10.1016/j.future.2020.06.045](https://doi.org/10.1016/j.future.2020.06.045)
- [14] Woodrow Barfield and Thomas A. Dingus. 2014. *Human Factors in Intelligent Transportation Systems*. Psychology Press.
- [15] Subhasree Basu, Yi Han Victoria Chua, Mei Wah Lee, Wanyu Geraldine Lim, Tomasz Maszczyk, Zheng Guo, and Justin Dauwels. 2018. Towards a data-driven behavioral approach to prediction of insider-threat. In *Proceedings of the 2018 IEEE International Conference on Big Data (Big Data)*. IEEE, 4994–5001.
- [16] Alison J. C. Bell, M. Brooke Rogers, and Julia M. Pearce. 2019. The insider threat: Behavioral indicators and factors influencing likelihood of intervention. *International Journal of Critical Infrastructure Protection* 24 (2019), 166–176. DOI: [10.1016/j.ijcip.2018.12.001](https://doi.org/10.1016/j.ijcip.2018.12.001)
- [17] Carson Brown, Alex Cowperthwaite, Abdulrahman Hijazi, and Anil Somayaji. 2009. Analysis of the 1999 DARPA/Lincoln laboratory IDS evaluation data with NetADHICT. In *Proceedings of the 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications*. IEEE, 1–7.
- [18] David P. Brown, Dennis Buade, and Sean D. Vermillion. 2019. Improving insider threat detection through multi-modelling/data fusion. *Procedia Computer Science* 153 (2019), 100–107. DOI: [10.1016/j.procs.2019.05.060](https://doi.org/10.1016/j.procs.2019.05.060)
- [19] Federal Infrastructure Protection Bureau. 2013. Unintentional insider threats: A foundational study. *Software Engineering Institute Technical Report* (2013).
- [20] Verizon Business. 2024. 2024 Data Breach Investigations Report. Retrieved June 1, 2024 from <https://www.verizon.com/business/resources/reports/2024-dbir-data-breach-investigations-report.pdf>
- [21] Matthew Canham, Clay Posey, and Patricia S. Bockelman. 2020. Confronting information security’s elephant, the unintentional insider threat. In *Augmented Cognition. Human Cognition and Behavior: 14th International Conference, AC 2020, Held as Part of the 22nd HCI International Conference, HCII 2020, Proceedings, Part II* 22. Springer, 316–334.
- [22] The CERT Insider Threat Center. 2013. Analyzing Insider Threat Data in the MERIT Database. Carnegie Mellon University, Software Engineering Institute’s Insights (blog). Retrieved October 19, 2024 from <https://insights.sei.cmu.edu/blog/analyzing-insider-threat-data-in-the-merit-database/>

- [23] Chia-Cheng Chen and Hsing-Kuo Pao. 2024. Reconsider time series analysis for insider threat detection. In *Proceedings of the 2024 IEEE International Conference on Big Data (BigData)*. IEEE, 1558–1565.
- [24] Ramkumar Chinchani, Duc Ha, Anusha Iyer, Hung Q. Ngo, and Shambhu Upadhyaya. 2010. Insider threat assessment: Model, analysis and tool. *Network Security* (2010), 143–174. DOI: [10.1007/978-0-387-73821-5_7](https://doi.org/10.1007/978-0-387-73821-5_7)
- [25] Matthew Collins, Michael Theis, Randall Trzeciak, Jeremy Strozer, Jason Clark, Daniel Costa, Tracy Cassidy, Michael Albrethsen, and Andrew Moore. 2016. Common sense guide to mitigating insider threats. *The CERT Insider Threat Center, Technical Report CMU/SEI-2015-TR-010*.
- [26] Daniel Costa. 2017. CERT Definition of 'Insider Threat' - Updated. Carnegie Mellon University, Software Engineering Institute's Insights (blog). Retrieved October 9, 2024 from <https://insights.sei.cmu.edu/blog/cert-definition-of-insider-threat-updated/>
- [27] Nitin Dixit, Rishi Gupta, and Pradeep Yadav. 2023. Insider threat classification using KNN machine-learning technique. In *Proceedings of the 2023 IEEE International Conference on Contemporary Computing and Communications (InC4)*, Vol. 1. IEEE, 1–5.
- [28] Jiankuo Dong, Jie Wei, Xiaoyu Hu, Zhenjiang Dong, Fuyuan Chen, Xin Hu, and Jin Qi. 2025. DDCC: Synergizing denoising diffusion probabilistic models and curriculum-based complexity control for insider threat detection. *IEEE Transactions on Industrial Informatics* 21, 11 (2025), 8351–8361. DOI: [10.1109/TII.2025.3574417](https://doi.org/10.1109/TII.2025.3574417)
- [29] Nebrase Elmrabit, Shuang-Hua Yang, Lili Yang, and Huiyu Zhou. 2020. Insider threat risk prediction based on Bayesian network. *Computers & Security* 96 (2020), 101908. DOI: [10.1016/j.cose.2020.101908](https://doi.org/10.1016/j.cose.2020.101908)
- [30] Arnau Erola, Ioannis Agraftiotis, Michael Goldsmith, and Sadie Creese. 2022. Insider-threat detection: Lessons from deploying the CITD tool in three multinational organisations. *Journal of Information Security and Applications* 67 (2022), 103167. DOI: <https://doi.org/10.1016/j.jisa.2022.103167>
- [31] Human Factors and Ergonomics Society. 2024. What Is Human Factors and Ergonomics? Retrieved May 10, 2024 from <https://www.hfes.org/About/What-Is-Human-Factors-and-Ergonomics>
- [32] Á Fülöp, L. Kovács, T. Kurics, and E. Windhager-Pokol. 2016. Balabit mouse dynamics challenge data set. Retrieved May 28, 2024 from b.com/balabit/Mouse-Dynamics-Challenge
- [33] Peng Gao, Haotian Zhang, Ming Wang, Weiyong Yang, Xinshen Wei, Zhuo Lv, and Zengzhou Ma. 2025. Deep temporal graph infomax for imbalanced insider threat detection. *Journal of Computer Information Systems* 65, 1 (2025), 108–118. DOI: <https://doi.org/10.1080/08874417.2023.2267510>
- [34] Haywood Gelman, John D. Hastings, and David Kenley. 2025. An ethically grounded LLM-based approach to insider threat synthesis and detection. (2025), 1–6. DOI: [10.1109/CARS67163.2025.11337953](https://doi.org/10.1109/CARS67163.2025.11337953)
- [35] Iffat A. Gheyas and Ali E. Abdallah. 2016. Detection and prediction of insider threats to cyber security: A systematic literature review and meta-analysis. *Big Data Analytics* 1, 1 (2016), 6.
- [36] Laurie Giddens, Laura C. Amo, and Dianna Cichocki. 2020. Gender bias and the impact on managerial evaluation of insider security threats. *Computers & Security* 99 (2020), 102066. DOI: [10.1016/j.cose.2020.102066](https://doi.org/10.1016/j.cose.2020.102066)
- [37] Raj Gopalakrishna and Rajaram Bhaskaran. 2025. Experience paper: Uncovering a privileged insider threat in action. In *Proceedings of the 2025 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*. IEEE, 656–661.
- [38] Anand K. Gramopadhye and Colin G. Drury. 2000. Human factors in aviation maintenance: How we got to where we are. *International Journal of Industrial Ergonomics* 26, 2 (2000), 125–131. DOI: [10.1016/S0169-8141\(99\)00062-1](https://doi.org/10.1016/S0169-8141(99)00062-1)
- [39] Marcus L. Green and Priya Dozier. 2023. Understanding human factors of cybersecurity: Drivers of insider threats. In *Proceedings of the 2023 IEEE International Conference on Cyber Security and Resilience (CSR)*. IEEE, 111–116.
- [40] Frank L. Greitzer. 2019. Insider threats: It's the HUMAN, stupid!. In *Proceedings of the Northwest Cybersecurity Symposium*. 1–8.
- [41] Frank L. Greitzer, Deborah Frincke, and Mariah Zabriskie. 2011. Social/ethical issues in predictive insider threat monitoring. In *Information Assurance and Security Ethics in Complex Systems: Interdisciplinary Perspectives*, Melissa Jane Dark (Ed.). IGI Global, 132–161. DOI: [10.4018/978-1-61692-245-0.ch007](https://doi.org/10.4018/978-1-61692-245-0.ch007)
- [42] Frank L. Greitzer, Jeremy Strozer, Sholom Cohen, John Bergey, Jennifer Cowley, Andrew Moore, and David Mundie. 2014. Unintentional insider threat: Contributing factors, observables, and mitigation strategies. In *Proceedings of the 2014 47th Hawaii International Conference on System Sciences*. IEEE, 2025–2034.
- [43] Frank L. Greitzer, Jeremy R. Strozer, Sholom Cohen, Andrew P. Moore, David Mundie, and Jennifer Cowley. 2014. Analysis of unintentional insider threats deriving from social engineering exploits. In *Proceedings of the 2014 IEEE Security and Privacy Workshops*. IEEE, 236–250.
- [44] Chathika Gunaratne, Mason Stott, Debraj De, and Gautam Malviya Thakur. 2024. MetaPoL: Immersive VR based indoor patterns of life (PoL) and Anomalies data generation for insider threat modeling in nuclear security. In *Proceedings of the 2024 25th IEEE International Conference on Mobile Data Management (MDM)*. IEEE, 255–258.
- [45] Lee Hadlington. 2021. The “human factor” in cybersecurity: Exploring the accidental insider. In *Research Anthology on Artificial Intelligence Applications in Security*, Information Resources Management Association (Ed.). IGI Global, 1960–1977. DOI: [10.4018/978-1-7998-7705-9.ch087](https://doi.org/10.4018/978-1-7998-7705-9.ch087)

- [46] Kholood Al tabash and Jassim Happa. 2018. Insider-threat detection using Gaussian mixture models and sensitivity profiles. *Computers & Security* 77 (2018), 838–859. DOI: [10.1016/j.cose.2018.03.006](https://doi.org/10.1016/j.cose.2018.03.006)
- [47] Athul Harilal, Flavio Toffalini, Ivan Homoliak, John Henry Castellanos, Juan Guarnizo, Soumik Mondal, and Martín Ochoa. 2018. The wolf Of SUTD (TWOS): A dataset of malicious insider threat behavior based on a gamified competition. *Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications* 9, 1 (2018), 54–85.
- [48] P. D. Harms, Alexander Marbut, Allen C. Johnston, Paul Lester, and Tyler Fezzey. 2022. Exposing the darkness within: A review of dark personality traits, models, and measures and their relationship to insider threats. *Journal of Information Security and Applications* 71 (2022), 103378. DOI: [10.1016/j.jisa.2022.103378](https://doi.org/10.1016/j.jisa.2022.103378)
- [49] Weiyu He, Xu Wu, Jingchen Wu, Xiaqing Xie, Lirong Qiu, and Lijuan Sun. 2021. Insider threat detection based on user historical behavior and attention mechanism. In *Proceedings of the 2021 IEEE 6th International Conference on Data Science in Cyberspace (DSC)*. IEEE, 564–569.
- [50] Diane Henshel, Char Sample, Mariana Cains, and Blaine Hoffman. 2016. Integrating cultural factors into human factors framework and ontology for cyber attackers. In *Advances in Human Factors in Cybersecurity: Proceedings of the AHFE 2016 International Conference on Human Factors in Cybersecurity, July 27-31, 2016, Walt Disney World®*. Springer, 123–137.
- [51] Isabel Herrera Montano, José Javier García Aranda, Juan Ramos Diaz, Sergio Molina Cardín, Isabel De la Torre Díez, and Joel JPC Rodrigues. 2022. Survey of techniques on data leakage protection and methods to address the insider threat. *Cluster Computing* 25, 6 (2022), 4289–4302.
- [52] Mils Hills and Anjali Anjali. 2017. A human factors contribution to countering insider threats: Practical prospects from a novel approach to warning and avoiding. *Security Journal* 30, 1 (2017), 142–152.
- [53] Ivan Homoliak, Flavio Toffalini, Juan Guarnizo, Yuval Elovici, and Martín Ochoa. 2019. Insight into insiders and it: A survey of insider threat taxonomies, analysis, modeling, and countermeasures. *ACM Computing Surveys (CSUR)* 52, 2 (2019), 1–40.
- [54] Wei Hong, Jiao Yin, Mingshan You, Hua Wang, Jinli Cao, Jianxin Li, Ming Liu, and Chengyuan Man. 2023. A graph empowered insider threat detection framework based on daily activities. *ISA Transactions* 141 (2023), 84–92. DOI: [10.1016/j.isatra.2023.06.030](https://doi.org/10.1016/j.isatra.2023.06.030)
- [55] Weiqing Huang, He Zhu, Ce Li, Qiujian Lv, Yan Wang, and Haitian Yang. 2021. Itdbert: Temporal-semantic representation for insider threat detection. In *Proceedings of the 2021 IEEE Symposium on Computers and Communications (ISCC)*. IEEE, 1–7.
- [56] Zilin Huang, Xiulai Li, Xinyi Cao, Ke Chen, Longjuan Wang, and Logan Bo-Yee Liu. 2024. IDU-Detector: A synergistic framework for robust masquerader attack detection. *IEEE Internet of Things Journal* 12, 8 (2024), 9653–9670. DOI: [10.1109/JIOT.2024.3503057](https://doi.org/10.1109/JIOT.2024.3503057)
- [57] Jeffrey Hunker and Christian W. Probst. 2011. Insiders and insider threats-an overview of definitions and mitigation techniques. *Journal of Wireless Mobile Networks, Ubiquitous Computing and Dependable Applications* 2, 1 (2011), 4–27.
- [58] IARPA. 2023. Scientific advances to Continuous Insider Threat Evaluation. Retrieved October 19, 2024 from <https://www.iarpa.gov/research-programs/scite>
- [59] Usman Inayat, Mashaim Farzan, Sajid Mahmood, Muhammad Fahad Zia, Shahid Hussain, and Fabiano Pallonetto. 2024. Insider threat mitigation: Systematic literature review. *Ain Shams Engineering Journal* 15, 12 (2024), 103068.
- [60] Faisal Janjua, Asif Masood, Haider Abbas, and Imran Rashid. 2020. Handling insider threat through supervised machine learning techniques. *Procedia Computer Science* 177 (2020), 64–71. DOI: [10.1016/j.procs.2020.10.012](https://doi.org/10.1016/j.procs.2020.10.012)
- [61] Myeongki Jeong and Hangjung Zo. 2021. Preventing insider threats to enhance organizational security: The role of opportunity-reducing techniques. *Telematics and Informatics* 63 (2021), 101670. DOI: [10.1016/j.tele.2021.101670](https://doi.org/10.1016/j.tele.2021.101670)
- [62] Bzhwen A. Kadir, Ole Broberg, and Carolina Souza da Conceição. 2019. Current research and future perspectives on human factors and ergonomics in Industry 4.0. *Computers & Industrial Engineering* 137 (2019), 106004. DOI: [10.1016/j.cie.2019.106004](https://doi.org/10.1016/j.cie.2019.106004)
- [63] K. Kamatchi and E. Uma. 2025. Insights into user behavioral-based insider threat detection: Systematic review. *International Journal of Information Security* 24, 2 (2025), 88.
- [64] Xiu Kan, Yixuan Fan, Jinjie Zheng, Chi-hung Chi, Wanqing Song, and Aleksey Kudreyko. 2023. Data adjusting strategy and optimized XGBoost algorithm for novel insider threat detection model. *Journal of the Franklin Institute* 360, 16 (2023), 11414–11443.
- [65] Neeshe Khan, Robert J. Houghton, and Sarah Sharples. 2022. Understanding factors that influence unintentional insider threat: A framework to counteract unintentional risks. *Cognition, Technology & Work* 24, 3 (2022), 393–421.
- [66] Aram Kim, Junhyoung Oh, Jinho Ryu, Jemin Lee, Kookheui Kwon, and Kyungho Lee. 2019. SoK: A systematic review of insider threat detection. *Journal of Wireless Mobile Networks, Ubiquitous Computing, and Dependable Applications* 10, 4 (2019), 46–67.
- [67] Aram Kim, Junhyoung Oh, Jinho Ryu, and Kyungho Lee. 2020. A review of insider threat detection approaches with IoT perspective. *IEEE Access* 8 (2020), 78847–78867. DOI: [10.1109/ACCESS.2020.2990195](https://doi.org/10.1109/ACCESS.2020.2990195)

- [68] Yujun Kim, Deok-Han Kim, and Young-Gab Kim. 2024. Transfer learning-based robust insider threat detection. In *Proceedings of the 2024 IEEE 23rd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*. IEEE, 2287–2292.
- [69] P. Lavanya, H. Anila Glory, and V. S. Shankar Sriram. 2024. Mitigating insider threat: A neural network approach for enhanced security. *IEEE Access* 12 (2024), 73752–73768. DOI : [10.1109/ACCESS.2024.3404814](https://doi.org/10.1109/ACCESS.2024.3404814)
- [70] P. Lavanya and V. S. Shankar Sriram. 2022. Detection of insider threats using deep learning: A review. In *Computational Intelligence in Data Mining: Proceedings of ICCIDM 2021*. 41–57.
- [71] Ximing Li, Linghui Li, Xiaoyong Li, Binsi Cai, Jia Jia, Yali Gao, and Shui Yu. 2024. GMFITD: Graph meta-learning for effective few-shot insider threat detection. *IEEE Transactions on Information Forensics and Security* 19 (2024), 7161–7175. DOI : [10.1109/TIFS.2024.3430106](https://doi.org/10.1109/TIFS.2024.3430106)
- [72] Yangyang Li and Yaping Su. 2023. The insider threat detection method of university website clusters based on machine learning. In *Proceedings of the 2023 6th International Conference on Artificial Intelligence and Big Data (ICAIBD)*. IEEE, 560–565.
- [73] Liu Liu, Olivier De Vel, Chao Chen, Jun Zhang, and Yang Xiang. 2018. Anomaly-based insider threat detection using deep autoencoders. In *Proceedings of the 2018 IEEE International Conference on Data Mining Workshops (ICDMW)*. IEEE, 39–48.
- [74] Jiuming Lu and Raymond K. Wong. 2019. Insider threat detection with long short-term memory. In *Proceedings of the Australasian Computer Science Week Multiconference*. 1–10.
- [75] Qiujian Lv, Yan Wang, Leiqi Wang, and Dan Wang. 2018. Towards a user and role-based behavior analysis method for insider threat detection. In *Proceedings of the 2018 International Conference on Network Infrastructure and Digital Content (IC-NIDC)*. IEEE, 6–10.
- [76] Qicheng Ma and Nidhi Rastogi. 2020. DANTE: Predicting insider threat using LSTM on system logs. In *Proceedings of the 2020 IEEE 19th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*. IEEE, 1151–1156.
- [77] Rachid Ait Maalem Lahcen, Bruce Caulkins, Ram Mohapatra, and Manish Kumar. 2020. Review and insight on the behavioral aspects of cybersecurity. *Cybersecurity* 3, 1 (2020), 1–18.
- [78] A. R. Marbut and P. D. Harms. 2024. Fiends and fools: A narrative review and neo-socioanalytic perspective on personality and insider threats. *Journal of Business and Psychology* 39, 3 (2024), 679–696. DOI : [10.1007/s10869-023-09885-9](https://doi.org/10.1007/s10869-023-09885-9)
- [79] Arshad Masood and Ammar Masood. 2021. A taxonomy of insider threat in isolated (air-gapped) computer networks. In *Proceedings of the 2021 International Bhurban Conference on Applied Sciences and Technologies (IBCAST)*. IEEE, 678–685.
- [80] Fanzhi Meng, Peng Lu, Junhao Li, Teng Hu, Mingyong Yin, and Fang Lou. 2021. GRU and Multi-autoencoder based insider threat detection for cyber security. In *Proceedings of the 2021 IEEE 6th International Conference on Data Science in Cyberspace (DSC)*. IEEE, 203–210.
- [81] Alexandra Ágnes Mészáros and Anikó Kelemen-Erdős. 2023. Industrial espionage from a human factor perspective. *Journal of International Studies (2071-8330)* 16, 3 (2023), 97–116. <https://www.cceol.com/search/article-detail?id=1192959>
- [82] Dietmar P. F. Möller. 2023. *Threats and Threat Intelligence*. Springer Nature Switzerland, Cham, 71–129. DOI : https://doi.org/10.1007/978-3-031-26845-8_2
- [83] Phillip L. Morgan, Phoebe M. Asquith, Laura M. Bishop, George Raywood-Burke, Adam Wedgbury, and Kevin Jones. 2020. A new hope: Human-centric cybersecurity research embedded within organizations. In *Proceedings of the International Conference on Human-Computer Interaction*. Springer, 206–216.
- [84] Lili Nemec Zlatolas, Tatjana Welzer, and Lenka Lhotska. 2024. Data breaches in healthcare: Security mechanisms for attack mitigation. *Cluster Computing* 27, 7 (2024), 8639–8654.
- [85] Jason R. C. Nurse, Philip A. Legg, Oliver Buckley, Ioannis Agrafiotis, Gordon Wright, Monica Whitty, David Upton, Michael Goldsmith, and Sadie Creese. 2014. A critical reflection on the threat from human insiders—its nature, industry perceptions, and detection approaches. In *Human Aspects of Information Security, Privacy, and Trust: 2nd International Conference, HAS 2014, Held as Part of HCI International 2014. Proceedings 2*. Springer, 270–281.
- [86] Tolulope O. Oladimeji, C. K. Ayo, and S. E. Adewumi. 2019. Review on insider threat detection techniques. *Journal of Physics: Conference Series* 1299, 1 (2019), 012046. DOI : [10.1088/1742-6596/1299/1/012046](https://doi.org/10.1088/1742-6596/1299/1/012046)
- [87] Keshnee Padayachee. 2022. An exploratory factor analysis of personality factors: An insider threat perspective. In *Proceedings of the International Symposium on Human Aspects of Information Security and Assurance*. Springer, 253–264.
- [88] Matthew J. Page, David Moher, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, Larissa Shamseer, Jennifer M. Tetzlaff, Elie A. Akl, Sue E. Brennan, et al. 2021. PRISMA 2020 explanation and elaboration: Updated guidance and exemplars for reporting systematic reviews. *BMJ* 372 (2021). DOI : <https://doi.org/10.1136/bmj.n160>

- [89] Preetam Pal, Pratik Chattopadhyay, and Mayank Swarnkar. 2023. Temporal feature aggregation with attention for insider threat detection from activity logs. *Expert Systems with Applications* 224 (2023), 119925. DOI : [10.1016/j.eswa.2023.119925](https://doi.org/10.1016/j.eswa.2023.119925)
- [90] Will Penny and R. Henson. 2007. Hierarchical models. *Statistical Parametric Mapping* (2007), 149–155. <https://www.fil.ion.ucl.ac.uk/~wpenny/publications/Ch13.pdf>
- [91] Shari Lawrence Pfleeger, Joel B. Predd, Jeffrey Hunker, and Carla Bulford. 2009. Insiders behaving badly: Addressing bad actors and their actions. *IEEE Transactions on Information Forensics and Security* 5, 1 (2009), 169–179.
- [92] Alessandro Pollini, Tiziana C. Callari, Alessandra Tedeschi, Daniele Ruscio, Luca Save, Franco Chiarugi, and Davide Guerri. 2022. Leveraging human factors in cybersecurity: An integrated methodological approach. *Cognition, Technology & Work* 24, 2 (2022), 371–390.
- [93] Sunitha Prabhu and Nik Thompson. 2020. A unified classification model of insider threats to information security. In *Proceedings of the 31st Australasian Conference on Information Systems*. DOI : <https://doi.org/20.500.11937/81763>
- [94] Sunitha Prabhu and Nik Thompson. 2022. A primer on insider threats in cybersecurity. *Information Security Journal: A Global Perspective* 31, 5 (2022), 602–611.
- [95] Badis Racherache, Paria Shirani, Andrei Soeanu, and Mourad Debbabi. 2023. CPID: Insider threat detection using profiling and cyber-persona identification. *Computers & Security* 132 (2023), 103350. DOI : [10.1016/j.cose.2023.103350](https://doi.org/10.1016/j.cose.2023.103350)
- [96] Krunal Randive, R Mohan, and Ambairam Muthu Sivakrishna. 2023. An efficient pattern-based approach for insider threat classification using the image-based feature representation. *Journal of Information Security and Applications* 73 (2023), 103434. DOI : [10.1016/j.jisa.2023.103434](https://doi.org/10.1016/j.jisa.2023.103434)
- [97] Usman Rauf, Mohamed Shehab, Nafees Qamar, and Sheema Sameen. 2021. Formal approach to thwart against insider attacks: A bio-inspired auto-resilient policy regulation framework. *Future Generation Computer Systems* 117 (2021), 412–425. DOI : [10.1016/j.future.2020.11.009](https://doi.org/10.1016/j.future.2020.11.009)
- [98] Usman Rauf, Zhiyuan Wei, and Fadi Mohsen. 2023. Employee watcher: A machine learning-based hybrid insider threat detection framework. In *Proceedings of the 2023 7th Cyber Security in Networking Conference (CSNet)*. IEEE, 39–45.
- [99] Madhu Raut, Sunita Dhavale, Amarjit Singh, and Atul Mehra. 2020. Insider threat detection using deep learning: A review. In *Proceedings of the 2020 3rd International Conference on Intelligent Sustainable Systems (ICISS)*. IEEE, 856–863.
- [100] Mark Rodbert. 2020. Why organisational readiness is vital in the fight against insider threats. *Network Security* 2020, 8 (2020), 7–9.
- [101] Krishna Chandra Roy and Guenevere Chen. 2024. GraphCH: A deep framework for assessing cyber-human aspects in insider threat detection. *IEEE Transactions on Dependable and Secure Computing* 21, 5 (2024), 4495–4509.
- [102] Asha S. and Shanmugapriya D. 2024. Understanding insiders in cloud adopted organizations: A survey on taxonomies, incident analysis, defensive solutions, challenges. *Future Generation Computer Systems* 158 (2024), 427–446. DOI : <https://doi.org/10.1016/j.future.2024.04.033>
- [103] Nader Sohrabi Safa, Carsten Maple, Steve Furnell, Muhammad Ajmal Azad, Charith Perera, Mohammad Dabbagh, and Mehdi Sookhak. 2019. Deterrence and prevention-based model to mitigate information security insider threats in organisations. *Future Generation Computer Systems* 97 (2019), 587–597. DOI : [10.1016/j.future.2019.03.024](https://doi.org/10.1016/j.future.2019.03.024)
- [104] Nader Sohrabi Safa, Carsten Maple, Tim Watson, and Rossouw Von Solms. 2018. Motivation and opportunity based model to reduce information security insider threats in organisations. *Journal of Information Security and Applications* 40 (2018), 247–257. DOI : [10.1016/j.jisa.2017.11.001](https://doi.org/10.1016/j.jisa.2017.11.001)
- [105] Malek Ben Salem and Salvatore J. Stolfo. 2011. Modeling user search behavior for masquerade detection. In *Proceedings of the International Workshop on Recent Advances in Intrusion Detection*. Springer, 181–200.
- [106] Neetesh Saxena, Emma Hayes, Elisa Bertino, Patrick Ojo, Kim-Kwang Raymond Choo, and Pete Burnap. 2020. Impact and key challenges of insider threats on organizations and critical businesses. *Electronics* 9, 9 (2020), 1460.
- [107] Jordan Richard Schoenherr. 2022. Insider threats and individual differences: Intention and unintentional motivations. *IEEE Transactions on Technology and Society* 3, 3 (2022), 175–184.
- [108] Jordan Richard Schoenherr and Robert Thomson. 2021. The cybersecurity (CSEC) questionnaire: Individual differences in unintentional insider threat behaviours. In *Proceedings of the 2021 International Conference on Cyber Situational Awareness, Data Analytics and Assessment (CyberSA)*. IEEE, 1–8.
- [109] Matthias Schonlau, William DuMouchel, Wen-Hua Ju, Alan F. Karr, Martin Theus, and Yehuda Vardi. 2001. Computer intrusion: Detecting masquerades. *Statistical Science* 16, 1 (2001), 58–74. <http://www.jstor.org/stable/2676780>
- [110] Sara Sinclair and Sean W. Smith. 2008. Preventative directions for insider threat mitigation via access control. In *Insider Attack and Cyber Security: Beyond the Hacker*, Salvatore J. Stolfo, Steven M. Bellovin, Angelos D. Keromytis, Shlomo Hershkop, Sean W. Smith, and Sinclair (Eds.). Springer, 165–194. DOI : [10.1007/978-0-387-77322-3_10](https://doi.org/10.1007/978-0-387-77322-3_10)
- [111] Charlie Soh, Sicheng Yu, Annamalai Narayanan, Santhiya Duraisamy, and Lihui Chen. 2019. Employee profiling via aspect-based sentiment and network for insider threats detection. *Expert Systems with Applications* 135 (2019), 351–361. DOI : [10.1016/j.eswa.2019.05.043](https://doi.org/10.1016/j.eswa.2019.05.043)

- [112] Statista. 2024. Retrieved February 1, 2024 from <https://business.sharpusa.com/simply-smarter-blog/to-err-is-human-unfortunately-5-key-findings-from-the-2023-verizon-data-breach-investigations-report>
- [113] Statista. 2024. Most significant cybersecurity threats in organizations worldwide according to Chief Information Security Officers (CISO) as of February 2023. Retrieved February 1, 2024 from <https://www.statista.com/statistics/1350460/cybersecurity-threats-at-companies-worldwide-cisos/>
- [114] Statista. 2024. Most significant cybersecurity threats in organizations worldwide according to Chief Information Security Officers (CISO) as of February 2023. Retrieved February 1, 2024 from [https://istari-global.com/insights/spotlight/2023-insider-threat-report/#:~:text=74%25%20of%20respondents%20feel%20moderately,disruption%20or%20outage%20\(41%25\).](https://istari-global.com/insights/spotlight/2023-insider-threat-report/#:~:text=74%25%20of%20respondents%20feel%20moderately,disruption%20or%20outage%20(41%25).)
- [115] Arshiya Subhani, Iftikhar Alam Khan, and Anmol Zubair. 2021. Review of insider and insider threat detection in the organizations. *Journal of Advanced Research in Social Sciences and Humanities* 6, 4 (2021), 167–174.
- [116] Xiaoshuang Sun, Yu Wang, and Zengkai Shi. 2021. Insider threat detection using an unsupervised learning method: COPOD. In *Proceedings of the 2021 International Conference on Communications, Information System and Computer Engineering (CISCE)*. IEEE, 749–754.
- [117] Tian Tian, Yan Zhu, Ning An, Bo Jiang, Huamin Feng, and Zhigang Lu. 2024. LSTM-Diff: A data generation method for imbalanced insider threat detection. In *Proceedings of the 2024 IEEE 23rd International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*. IEEE, 68–76.
- [118] Leela Pavani Velagala and Gahangir Hossain. 2023. Analyzing insider threats and human factors in healthcare 5.0. In *Proceedings of the 2023 IEEE 20th International Conference on Smart Communities: Improving Quality of Life using AI, Robotics and IoT (HONET)*. IEEE, 95–100.
- [119] David S. Wall. 2013. Enemies within: Redefining the insider threat in organizational security policy. *Security Journal* 26, 2 (2013), 107–124. DOI : [10.1057/sj.2012.1](https://doi.org/10.1057/sj.2012.1)
- [120] Jiarong Wang, Qianran Sun, and Caiqiu Zhou. 2023. Insider threat detection based on deep clustering of multi-source behavioral events. *Applied Sciences* 13, 24 (2023), 13021.
- [121] Shannon Wasko, Rebecca E. Rhodes, Megan Goforth, Nathan Bos, Hannah P. Cowley, Gerald Matthews, Alice Leung, Satish Iyengar, and Jonathon Kopecky. 2021. Using alternate reality games to find a needle in a haystack: An approach for testing insider threat detection methods. *Computers & Security* 107 (2021), 102314. DOI : [10.1016/j.cose.2021.102314](https://doi.org/10.1016/j.cose.2021.102314)
- [122] Patrick Waterson and Ken Catchpole. 2016. Human factors in healthcare: Welcome progress, but still scratching the surface. *BMJ Quality & Safety* 25, 7 (2016), 480–484.
- [123] Findlay Whitelaw, Jackie Riley, and Nebrase Elmrabit. 2024. A review of the insider threat, a practitioner perspective within the UK financial services. *IEEE Access* 12 (2024), 34752–34768. DOI : [10.1109/ACCESS.2024.3373265](https://doi.org/10.1109/ACCESS.2024.3373265)
- [124] Wai Peng Wong, Hwee Chin Tan, Kim Hua Tan, and Ming-Lang Tseng. 2019. Human factors in information leakage: Mitigation strategies for information sharing integrity. *Industrial Management & Data Systems* 119, 6 (2019), 1242–1267.
- [125] Fengrui Xiao, Shuangwu Chen, Siyang Chen, Yuanyi Ma, Huasen He, and Jian Yang. 2024. SENTINEL: Insider threat detection based on multi-timescale user behavior interaction graph learning. *IEEE Transactions on Network Science and Engineering* 12, 2 (2024), 774–790. DOI : [10.1109/TNSE.2024.3519155](https://doi.org/10.1109/TNSE.2024.3519155)
- [126] Haitao Xiao, Yan Zhu, Bin Zhang, Zhigang Lu, Dan Du, and Yuling Liu. 2024. Unveiling shadows: A comprehensive framework for insider threat detection based on statistical and sequential analysis. *Computers & Security* 138 (2024), 103665. DOI : [10.1016/j.cose.2023.103665](https://doi.org/10.1016/j.cose.2023.103665)
- [127] Xiaoyun Ye, Huangrongbin Cui, Faqin Luo, Jinlong Wang, Xiaoyun Xiong, Wencui Zhang, Jiawei Yu, and Wenhao Zhao. 2025. Daily insider threat detection with hybrid TCN transformer architecture. *Scientific Reports* 15, 1 (2025), 28590.
- [128] Heather Young, Tony van Vliet, Josine van de Ven, Steven Jol, and Carlijn Broekman. 2018. Understanding human factors in cyber security as a dynamic system. In *Advances in Human Factors in Cybersecurity: Proceedings of the AHFE 2017 International Conference on Human Factors in Cybersecurity*. Springer, 244–254.
- [129] Fangfang Yuan, Yanmin Shang, Yanbing Liu, Yanan Cao, and Jianlong Tan. 2020. Data augmentation for insider threat detection with GAN. In *Proceedings of the 2020 IEEE 32nd International Conference on Tools with Artificial Intelligence (ICTAI)*. IEEE, 632–638.
- [130] Shuhan Yuan and Xintao Wu. 2021. Deep learning for insider threat detection: Review, challenges and opportunities. *Computers & Security* 104 (2021), 102221. DOI : [10.1016/j.cose.2021.102221](https://doi.org/10.1016/j.cose.2021.102221)
- [131] Shuhan Yuan, Panpan Zheng, Xintao Wu, and Qinghua Li. 2019. Insider threat detection via hierarchical neural temporal point processes. In *Proceedings of the 2019 IEEE International Conference on Big Data (Big Data)*. IEEE, 1343–1350.
- [132] Chaofan Zheng, Wenhui Hu, Tianci Li, Xueyang Liu, Jinchuan Zhang, and Litian Wang. 2022. An insider threat detection method based on heterogeneous graph embedding. In *Proceedings of the 2022 IEEE 8th Intl Conference on Big Data Security on Cloud (BigDataSecurity), IEEE Intl Conference on High Performance and Smart Computing (HPSC) and IEEE Intl Conference on Intelligent Data and Security (IDS)*. IEEE, 11–16.

[133] Xingjian Zhu, Jiankuo Dong, Jin Qi, Zhenguo Zhou, Zhenjiang Dong, Yanfei Sun, and Moyu Wang. 2024. AUTH: An adversarial autoencoder based unsupervised insider threat detection scheme for multisource logs. *IEEE Transactions on Industrial Informatics* 20, 9 (2024), 10954–10965.

A Appendix

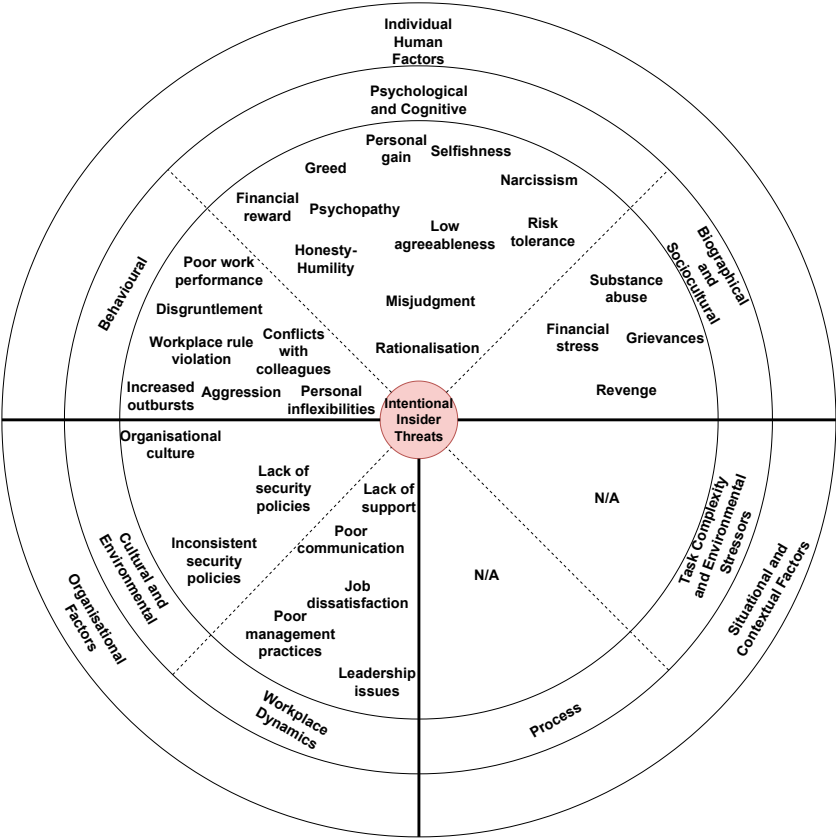


Fig. 4. Human factors contributing to intentional insider threats.

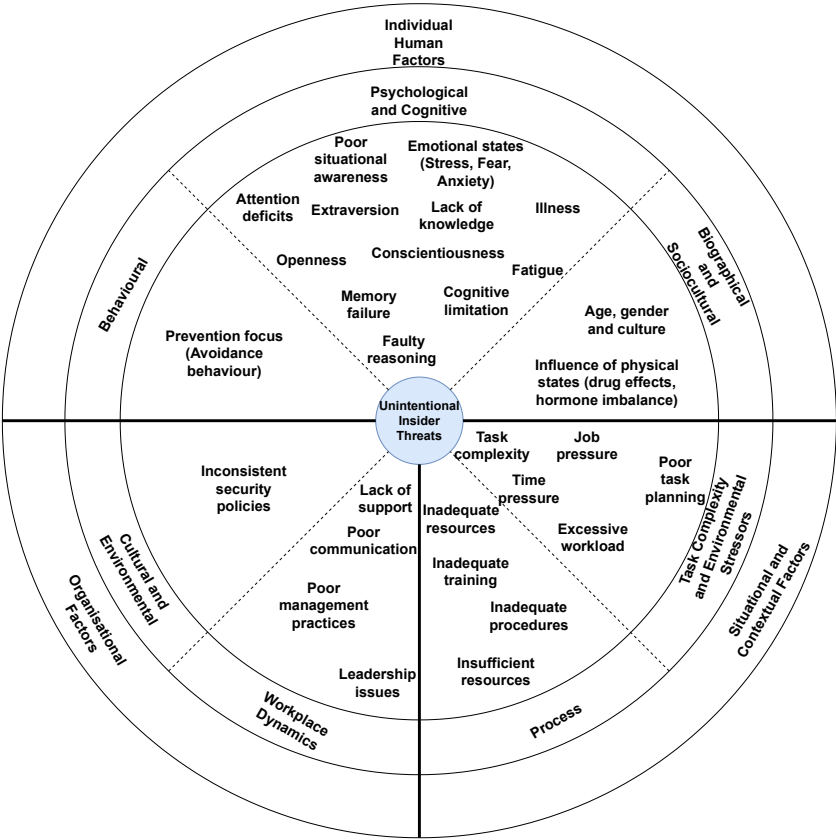


Fig. 5. Human factors contributing to unintentional insider threats.

Received 19 December 2024; revised 27 November 2025; accepted 4 February 2026